

ON THE ELLIPTIC HALL CATEGORY

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ABSTRACT. In this paper we begin to study categorical versions of (the positive half of) the elliptic Hall category, a categorical version of the elliptic Hall algebra. Our main result is a computation of categorical versions of the generators in the coherent realization of the elliptic Hall category, i.e., via coherent sheaves on commuting stacks for $GL_n \times \mathfrak{gl}_n$: we realize them as Hodge associated graded of certain twisted equivariant D -modules on the groups GL_n .

1. INTRODUCTION

This paper deals with an *elliptic Hall category*, a categorical version of the elliptic Hall algebra. The latter has been studied quite extensively in the last 30 years or so. One exciting feature of this algebra is that it admits several different realizations:

- (i) As a *Ding-Iohara algebra*, a quantum group for the double Heisenberg algebra associated to $\mathfrak{gl}_1$, [DI].
- (ii) As a suitable limit as $n \rightarrow \infty$ of spherical double affine Hecke algebras for GL_n , [SV1].
- (iii) As a suitable version of the Hall algebra of an elliptic curve over a finite field, [BS].
- (iv) As a sort of a double of the direct sum (over $n \geq 0$) of equivariant K-theories of commuting stacks for the Lie algebras $\mathfrak{gl}_n$, [SV2].

Let us review a few features of this algebra. First, it is an algebra over $\mathbb{C}(q, t)$. Second, it has generators labeled by $\mathbb{Z}^2 \setminus \{(0, 0)\}$. From here we get a triangular decomposition, where the positive part corresponds to the generators labeled by $(r, a) \in \mathbb{Z}^2$ with $r > 0$. For example, in realization (iv), the positive part is the direct sum of the equivariant K-theories of commuting stacks for $\mathfrak{gl}_n$ as in (iv) with $r = n$. A natural question therefore is to categorify the elliptic Hall algebra or its positive part (one could hope to recover a categorification of the former from that of the latter, compare with Kac-Moody categorifications).

We note that realizations (iii) and (iv) above suggest categorifications of the positive part: (iii) in terms of constructible sheaves on moduli spaces of vector bundles on the elliptic curves, while (iv) in terms of coherent sheaves on commuting stacks on $\mathfrak{gl}_n$'s. One expects that the resulting categorifications are equivalent, this equivalence can be thought of as a double affine relative of an equivalence of two categorifications of the affine Hecke algebra by the first named author, [B].

In fact, there is the third candidate for being a categorification of the positive part of the elliptic Hall algebra coming from the trace categories of affine Hecke categories for GL_n that we now describe.

Assume for now that G is an arbitrary connected reductive group and let G/B be its flag variety. We write $\mathcal{Z}$ for the Steinberg dg-scheme, i.e., the derived fiber product $T^*(G/B) \times_{\mathfrak{g}^*}^R T^*(G/B)$, where the map $T^*(G/B) \rightarrow \mathfrak{g}^*$ is the moment map for the action of G . We consider the derived category of $G \times \mathbb{C}^\times$ -equivariant coherent sheaves on $\mathcal{Z}$, this is the affine Hecke

category for G to be denoted by $\mathcal{AH}$. This category is monoidal with respect to convolution. In particular, we consider it trace category, cf. [BNP]. In [BNP, Theorem 1.4.8], the authors identify this trace category with the derived category of $G \times \mathbb{C}^\times$ -equivariant coherent sheaves on the derived scheme $\mathfrak{C} := \{(g, x) \in G \times \mathfrak{g} \mid \text{Ad}(g)x = x\}$ set-theoretically supported on the locus (g, x) with nilpotent x . The latter category will be denoted by $\text{Coh}_{\mathcal{N}, G \times \mathbb{C}^\times}(\mathfrak{C})$. The trace functor $\mathcal{AH} \rightarrow \text{Coh}_{\mathcal{N}, G \times \mathbb{C}^\times}(\mathfrak{C})$ is the usual pull-push functor between coherent sheaves on cotangent bundles using the correspondence $B \backslash G/B \leftarrow G/\text{Ad}(B) \rightarrow G/\text{Ad}(G)$, we will recall the construction in more detail in Section 2.6. Denote this functor by $\widetilde{\text{CH}}$. In this paper we will only use this functor but not the claim of [BNP, Theorem 1.4.8].

Assume now that $G = \text{GL}_n$. We decorate the schemes and categories appearing in the previous paragraph for $G = G_n := \text{GL}_n$ with superscript n : $\mathcal{AH}_n, \mathfrak{C}_n$, etc. For $m, n \in \mathbb{Z}_{\geq 0}$, let $P_{m,n} \subset G_{m+n}$ denote the stabilizer of $\mathbb{C}^m \subset \mathbb{C}^{m+n}$. Thanks to the homomorphisms $G_m \times G_n \leftarrow P_{m,n} \hookrightarrow G_{m+n}$, we get functors (to be called *parabolic induction*)

$$\begin{aligned} \mathcal{AH}_m \boxtimes \mathcal{AH}_n &\rightarrow \mathcal{AH}_{m+n}, \\ \text{Coh}_{\mathcal{N}, G_m \times \mathbb{C}^\times}(\mathfrak{C}_m) \boxtimes \text{Coh}_{\mathcal{N}, G_n \times \mathbb{C}^\times}(\mathfrak{C}_n) &\rightarrow \text{Coh}_{\mathcal{N}, G_{m+n} \times \mathbb{C}^\times}(\mathfrak{C}_{m+n}). \end{aligned}$$

It is not difficult to see (cf. Section 5) that the parabolic induction functors intertwine $\widetilde{\text{CH}}_m \boxtimes \widetilde{\text{CH}}_n$ with $\widetilde{\text{CH}}_{m+n}$. Both

$$\bigoplus_{n \geq 0} \text{Tr}(\mathcal{AH}_n), \bigoplus_{n \geq 0} \text{Coh}_{\mathcal{N}, G_n \times \mathbb{C}^\times}(\mathfrak{C}_n)$$

acquire monoidal structure via parabolic induction and $\bigoplus_{n \geq 0} \widetilde{\text{CH}}_n$ becomes a monoidal equivalence.

Now we proceed to describing the main result of our paper. Recall that the elliptic Hall algebra comes with distinguished generators labeled by pairs of integers $(r, a) \neq (0, 0)$. For the positive half we take $r > 0$. A natural question is to determine natural categorifications of these elements and find their preimages under the functor $\widetilde{\text{CH}}_r$. A closely related question was asked by Gorsky and Negut, [GN1, GN2] in a somewhat different setting, we'll discuss a relation between their work and ours below in this section and in Section 6.2.

Note that the subgroup of length 0 elements in the affine Weyl group for GL_r is identified with $\mathbb{Z}$, let ϖ denote a generator. An expected categorification of the generator corresponding to (r, a) is $\widetilde{\text{CH}}(\varpi^a)$ at least when r, a are coprime. The goal of this paper is to compute this coherent sheaf on the commuting scheme $\mathfrak{C}_r$. Let us explain this result here.

Assume that r and a are coprime. One has the unique irreducible twisted GL_r -equivariant irreducible D-module on the unipotent cone of PGL_r , where the twist is by the character $\frac{a}{r} \text{tr}$ of $\mathfrak{gl}_r$. Let $\mathbb{N}_{a,r}$ denote the pullback of this D-module to GL_r . The D-module $\mathbb{N}_{a,r}$ comes with a mixed Hodge structure, hence with a distinguished choice of a good filtration. Taking the associated graded we, in fact, get a $\text{GL}_r \times \mathbb{C}^\times$ -equivariant coherent sheaf on $\mathfrak{C}_r$. We show that this equivariant sheaf is isomorphic to $\widetilde{\text{CH}}(\varpi^a)$. The proof exploits the compatibility between the operation of taking the Hodge associated graded and the functor $\widetilde{\text{CH}}$. Namely, we identify an object in a suitable finite Hecke category whose Hodge associated graded sheaf coincides with ϖ^a , and show that the finite analog of the character functor maps this object essentially to $\mathbb{N}_{a,r}$.

In the preceding discussion we assumed that r, a are coprime. We have not computed the image of ϖ^a for general a . However, from some perspective, this is not a natural question. Instead, one uses the following construction. Let $d = \text{GCD}(r, a)$, $r_0 = r/d$, $a_0 = a/d$, so

that (r_0, a_0) are coprime. Instead of considering $\widetilde{\mathrm{CH}}(\varpi^a)$, one can try to construct an action of S_d on $\widetilde{\mathrm{CH}}(\varpi^{a_0})^d$, where the power is for the monoidal structure coming from parabolic induction. Then one looks at the S_d -isotypic components, the object $\widetilde{\mathrm{CH}}(\varpi^a)$ should be a complex of the isotypic components. In the main body of the paper we indeed construct an S_d -action, again using a connection to Hodge D-modules.

Remark 1.1. Here is one reason an equivalence $\mathrm{Tr}(\mathcal{AH}) \cong \mathrm{Coh}_{\mathcal{N}, G \times \mathbb{C}^\times}(\mathfrak{C})$ is useful: the left hand side has a conjectural realization in terms of constructible sheaves, [LNY, Conjecture 1.5.1]. So, modulo this conjecture, we get an equivalence of categorical versions of the positive halves of the elliptic Hall algebras in realizations (iii) and (iv) above. This equivalence can be viewed as a double affine version of the main equivalence in [B]. It should be an ingredient in a constructible realization of a category of modular representations of rational Cherednik algebras associated to the symmetric groups.

Finally, we would like to discuss a connection between our work and [GN1, GN2]. The purpose of these papers is to relate the cocenters of the affine Hecke categories for GL_n to the category of $(\mathbb{C}^\times)^2$ -equivariant coherent sheaves on the additive version of the commuting stacks: $\{(x, y) \in \mathfrak{gl}_n \mid [x, y] = 0\} / \mathrm{GL}_n$. Here the two copies of $(\mathbb{C}^\times)^2$ act by rescaling x, y individually. This, still conjectural, realization has the advantage to directly decategorify (by passing to the Grothendieck groups) the positive half of the elliptic Hall algebra that is missing in our approach. On the other hand, our approach is more natural from the geometric and representation theoretic perspectives. It is also worth mentioning that our answer for $\widetilde{\mathrm{CH}}(\varpi^a)$ has a direct additive analog, one can, in fact, show that, in the coprime case the additive analogs of $\widetilde{\mathrm{CH}}(\varpi^a)$ defined as Hodge associated graded of certain twisted equivariant D-modules on $\mathfrak{gl}_r$ coincide with the sheaves $\mathcal{H}'_{a,r}$ from [GN2, Section 2.3].

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2. PRELIMINARIES

2.1. Equivariant mixed Hodge modules. In Saito's theory [Sai, Sai2], the structure of a (polarizable) mixed Hodge module on a smooth algebraic variety X consists of a regular holonomic left D_X -module, a weight filtration, a good filtration on M called the Hodge filtration, a polarization, and a $\mathbb{Q}$ -structure satisfying compatibility conditions. We use the complex version in the sense of Sabbah and Schnell [SS], in which the $\mathbb{Q}$ -structure is omitted. Polarizations will not play a role for our purposes either.

All functors below are derived unless otherwise specified. Let $f : X \rightarrow Y$ be a morphism between two smooth varieties, we reserve f_* and f^* for pushforward and pullback functors for quasi-coherent sheaves. Suppose the smooth variety X has an action of an affine algebraic group G . Let $D_G^b\mathrm{MHM}(X)$ be the equivariant derived category of mixed Hodge modules on X as in [A]. These derived categories admit the six functor formalism. Let $\mathcal{D}_{X \rightarrow Y}$ and $\mathcal{D}_{Y \leftarrow X}$ be the transfer D-modules as defined in [HTT, 1.3]. We write $f_{\dagger} := f_*(\mathcal{D}_{Y \leftarrow X} \otimes_{\mathcal{D}_X} -)$. Let $\mathbb{D}_X = \mathrm{Hom}_{\mathcal{D}_X}(-, \mathcal{D}_X)[\dim(X)]$ be the Verdier duality. Let $f_{\dagger} = \mathbb{D}_Y^{-1} \circ f_{\dagger} \circ \mathbb{D}_X$. Define $f^{\dagger} = \mathcal{D}_{X \rightarrow Y} \otimes_{f^*\mathcal{D}_Y} f^*(-)$ and $f^{\dagger} = \mathbb{D}_X^{-1} \circ f^{\dagger} \circ \mathbb{D}_Y$.

For a $\mathbb{C}^\times$ -equivariant sheaf $\mathcal{F}$, or equivalently a grading $\mathcal{F} = \bigoplus_{k \in \mathbb{Z}} \mathcal{F}_k$, let $\mathcal{F}\langle \ell \rangle$ denotes the shift such that $(\mathcal{F}\langle \ell \rangle)_k = \mathcal{F}_{k-\ell}$.

2.2. Associated graded of direct/inverse images. For any morphism $f : X \rightarrow Y$ between two smooth varieties, we introduce the associated Lagrangian correspondence as indicated by (1): the map ρ_f is the co-differential of f and ϖ_f is the natural projection.

$$(1) \quad T^*X \xleftarrow{\rho_f} X \times_Y T^*Y \xrightarrow{\varpi_f} T^*Y.$$

Let $\tilde{\omega}_{X/Y}$ denote the inverse image of the relative canonical bundle $\omega_{X/Y}$ to $X \times_Y T^*Y$.

Theorem 2.1. ([Lau, 2.3.2], [Sch, Section 28]) *Let*

$$R\gamma_p(-) := R(\varpi_p)_*(L\rho_p^*(-) \otimes \tilde{\omega}_{X/Y}\langle \dim(Y) - \dim(X) \rangle).$$

When $p : X \rightarrow Y$ is a projective morphism, the following diagram commutes.

$$\begin{array}{ccc} D_G^b\text{MHM}(X) & \xrightarrow{p_!} & D_G^b\text{MHM}(Y) \\ \downarrow \text{gr} & & \downarrow \text{gr} \\ \text{Coh}_G(T^*X) & \xrightarrow{\gamma_p} & \text{Coh}_G(T^*Y) \end{array}$$

Moreover, the direct image is strict, i.e.,

$$\text{gr}^F(R^i p_! M) \cong R^i \gamma_p(\text{gr}^F M).$$

Lemma 2.2. ([Sch, Theorem 30.1], [Ka, Theorem 4.7]) *Suppose $f : X \rightarrow Y$ is a smooth morphism. The following diagram commutes.*

$$\begin{array}{ccc} D_G^b\text{MHM}^G(Y) & \xrightarrow{f^\dagger} & D_G^b\text{MHM}^G(X) \\ \downarrow \text{gr} & & \downarrow \text{gr} \\ \text{Coh}_G(T^*Y) & \xrightarrow{(\rho_f)_* \varpi_f^*} & \text{Coh}_G(T^*X) \end{array}$$

2.2.1. Inductions. Let H be a closed subgroup of G , which acts on the product $G \times X$ by $h \cdot (g, x) = (gh^{-1}, hx)$. Let $G \times^H X$ be the quotient of this action. Consider the diagram:

$$X \xleftarrow{pr} G \times X \xrightarrow{\pi} G \times^H X$$

where pr is the second projection, π is the quotient map.

For any H -equivariant $\mathcal{D}_X$ -module $\mathbb{F}$, there is a unique G -equivariant $\mathcal{D}_{G \times^H X}$ -module $\mathbb{E}$ such that $pr^\dagger \mathbb{F} \cong \pi^! \mathbb{E}$ [BeLu]. This functor lifts to an equivalence $\text{HM}^H(X) \xrightarrow{\sim} \text{HM}^G(G \times^H X)$. We denote $\mathbb{E} = \text{Ind}_H^G \mathbb{F}$. Let $\mu_H : T^*X \rightarrow \text{Lie}(H)^*$ be the moment map. Strong H -equivariance implies that $\text{gr}(\mathbb{F})$ is naturally an H -equivariant coherent sheaf on $\mu_H^{-1}(0)$. For any H -equivariant coherent $\mathcal{O}_X$ -module $\mathbb{F}$ on $\mu_H^{-1}(0)$, we write $\text{ind}_H^G \mathbb{F}$ to denote the corresponding object in $\text{Coh}_G(G \times^H \mu_H^{-1}(0))$. Let s denote the natural closed embedding $G \times^H \mu_H^{-1}(0) \hookrightarrow T^*(G \times^H X)$.

Proposition 2.3. [A, Theorem 6.2][Ma, Proposition 3.4] *We have the following commutative diagram.*

$$\begin{array}{ccc} \mathrm{HM}^H(X) & \xrightarrow[\sim]{\mathrm{Ind}_H^G} & \mathrm{HM}^G(G \times^H X) \\ \downarrow \tilde{\mathrm{gr}} & & \downarrow \tilde{\mathrm{gr}} \\ \mathrm{Coh}_H(\mu_H^{-1}(0)) & \xrightarrow[\sim]{s_* \mathrm{ind}_H^G} & \mathrm{Coh}_G(s(G \times^H \mu_H^{-1}(0))). \end{array}$$

2.3. Monodromic Hecke categories. Let G be a complex connected reductive group with Lie algebra $\mathfrak{g}$. Inside G , we fix a Cartan subgroup and a Borel subgroup $T \subset B$, with Lie algebras $\mathfrak{t} \subset \mathfrak{b}$. Let $U = [B, B]$ be the unipotent radical of B with Lie algebra $\mathfrak{n}$. Denote the flag variety G/B by $\mathcal{B}$ and the basic affine space G/U by $\tilde{\mathcal{B}}$.

Consider the right diagonal action of T on $\tilde{\mathcal{B}} \times \tilde{\mathcal{B}}$ and define $Y := (\tilde{\mathcal{B}} \times \tilde{\mathcal{B}})/T$. On Y we have the diagonal left G -action and a T -action by $t(x, y) = (x, yt^{-1})$. The fibration $\tilde{\mathcal{B}}^2 \rightarrow Y$ is G -equivariant.

We follow the definitions in [LY]. Let $m : T \times T \rightarrow T$ be the multiplication map. A rank one character D -module on T is a rank 1 regular integrable connection equipped with an isomorphism $m^* \mathcal{L} \cong \mathcal{L} \boxtimes \mathcal{L}$ and a trivialization of the stalk at e . The isomorphism classes of all rank one character D -modules form a group, which we denote by $Ch(T)$. Suppose $\mathcal{L}$ is of finite order, or equivalently, the character $\lambda \in \mathfrak{t}^*$ that determines the integrable connection is rational. Then there is a finite étale cover

$$\alpha : \tilde{T} \rightarrow T$$

such that $\mathcal{L} = (\alpha_* \mathcal{O}_{\tilde{T}})^\theta$ with respect to $\theta = e^{2\pi i \lambda} : \mathrm{Ker}(\alpha) \rightarrow \mathbb{C}^\times$. There is an action of W on $Ch(T)$ given by $w\mathcal{L} = (w^{-1})^* \mathcal{L}$. Let $W_\theta \subset W$ be the stabilizer of $\mathcal{L}$, or equivalently the stabilizer of θ .

Suppose a smooth variety X has a T action, and in particular a $\tilde{T}$ action through the cover α . Under the action of $\mathrm{Ker}(\alpha)$, $D_{\tilde{T}}^b \mathrm{MHM}(X)$ decomposes into a direct sum of triangulated subcategories on which $\mathrm{Ker}(\alpha)$ acts by characters. Let $D_\theta^b \mathrm{MHM}(X)$ denote the direct summand associated to θ . When $\theta = 1$, we recover $D_T^b \mathrm{MHM}(X)$.

We further impose G -equivariance and set $\mathcal{H}_\theta := D_{G, \theta}^b \mathrm{MHM}(Y)$; its heart is denoted by $\mathcal{H}_\theta^\heartsuit$.

On $\tilde{\mathcal{B}}$, we have the adjoint B -action. Consider the quotient $G \times^B \tilde{\mathcal{B}}$ of $G \times \tilde{\mathcal{B}}$ by the diagonal B -action:

$$b(g, x) = (gb^{-1}, bxb^{-1}), g \in G, x \in \tilde{\mathcal{B}}.$$

On $G \times^B \tilde{\mathcal{B}}$, we have a $G \times T$ -action defined by $(h, t)(g, x) = (hg, xt^{-1})$. Suppose $g \in G, x \in \tilde{\mathcal{B}}$. The map

$$G \times^B \tilde{\mathcal{B}} \xrightarrow{\sim} Y, \quad (g, x) \mapsto (g, gx)$$

defines a $G \times T$ -equivariant isomorphism.

The $G \times T$ -orbits in Y are in bijection with the $B \times T$ -orbits in $\tilde{\mathcal{B}}$ and the Weyl group W : for every $w \in W$ denote its lift in $N(T)$ still by w . Write $\mathcal{B}_w = BwB/B$ and $\tilde{\mathcal{B}}_w = BwB/U$.

Let U_- be the opposite of U . Take $U_w = U \cap w \cdot U_-$. The embedding $U_w wT \rightarrow \tilde{\mathcal{B}}_w$ is an isomorphism, and so finite-rank regular integrable connections on $\tilde{\mathcal{B}}_w$ are in bijection with finite-dimensional representations of $\pi_1(T)$.

Let $\Gamma_w = \{(t, w^{-1}(t)t^{-1}) \in T \times T \mid t \in T\}$ and $U^w = U \cap w \cdot U$. For any point $x \in \tilde{\mathcal{B}}_w$, the stabilizer in $\mathrm{Ad}(B) \times T$ is isomorphic to $\Gamma_w \ltimes U^w$.

Let $i_w : \{w\} \hookrightarrow \tilde{\mathcal{B}}_w$. Similar to [LY, Lemma 2.10], we have the following lemma.

Lemma 2.4. *The category $D^b\mathrm{MHM}_{B,\theta}(\tilde{\mathcal{B}}_w)$ is nonzero if and only if $w \in W_\theta$. In this case, we have an equivalence*

$$i_w^\dagger[-\dim \tilde{\mathcal{B}}_w] : D_{B,\theta}^b\mathrm{MHM}(\tilde{\mathcal{B}}_w) \xrightarrow{\sim} D_{\Gamma_w}^b\mathrm{MHM}(\{w\}).$$

Let

$$(2) \quad \mathbb{L}_{w,\theta} \in D_{B,\theta}^b\mathrm{MHM}(\tilde{\mathcal{B}}_w)$$

be the rank-1 object corresponding to $\mathbb{C}_{\{w\}}$ of type $(0, 0)$. Let $j : \tilde{\mathcal{B}}_w \rightarrow \tilde{\mathcal{B}}$ be the embedding. Define

$$(3) \quad \nabla_{w,\theta}^e = j_! \mathbb{L}_{w,\theta} \quad \nabla_{w,\theta} = \mathrm{Ind}_B^G \nabla_{w,\theta}^e;$$

$$(4) \quad \Delta_{w,\theta}^e = j_! \mathbb{L}_{w,\theta} \quad \Delta_{w,\theta} = \mathrm{Ind}_B^G \Delta_{w,\theta}^e;$$

$$IC_{w,\theta}^e = j_! \mathbb{L}_{w,\theta} \quad IC_{w,\theta} = \mathrm{Ind}_B^G IC_{w,\theta}^e;$$

where the objects in the second column are in $\mathcal{H}_\theta^\vee$. Denote $\nabla_w^e = \nabla_{w,1}^e$ and $\nabla_w = \nabla_{w,1}$, etc. We call the objects $\nabla_{w,\theta}$ (resp., $\Delta_{w,\theta}$) costandard (resp., standard) objects: one can show that they are the standard and costandard objects for a highest weight structure on $\mathcal{H}_\theta^\vee$ but we are not going to use this.

2.4. Character sheaves. There is the horocycle correspondence:

$$(5) \quad G \xleftarrow{p} G \times \mathcal{B} \xrightarrow{a} Y$$

where $a : (g, h) \mapsto \overline{(\tilde{h}, g\tilde{h})}$ with $\tilde{h}$ being a lift of $h \in \mathcal{B}$ in $\tilde{\mathcal{B}}$ and $p : (g, h) \mapsto g$.

Consider the functor

$$\mathrm{CH} := p_! a^\dagger : \mathcal{H}_\theta \rightarrow D_G^b\mathrm{MHM}(G).$$

Define $\mathcal{CS}_\theta^\vee \subset \mathrm{MHM}_G(G)$ as the smallest Serre subcategory containing the objects $\mathcal{H}^i \mathrm{CH}(\mathcal{F})$, where $\mathcal{F} \in \mathcal{H}_\theta^\vee$ and $i \in \mathbb{Z}$. Further, let $\mathcal{CS}_\theta \subset D_G^b\mathrm{MHM}(G)$ denote the full subcategory consisting of complexes all of whose cohomology objects belong to $\mathcal{CS}_\theta^\vee$. Since $\mathcal{CS}_\theta^\vee$ is a Serre subcategory, the long exact cohomology sequence shows that $\mathcal{CS}_\theta$ is triangulated; its definition in terms of cohomology objects also makes it stable under the standard truncation functors. Its heart is precisely $\mathcal{CS}_\theta^\vee$.

Remark 2.5. Applying these results to the elliptic Hall category (where the relevant groups are $G = \mathrm{GL}_n$) requires a monodromic setting. Let $\chi : G \rightarrow \mathbb{C}^\times$ be a rational character. Choose a central isogeny $q : \tilde{G} \rightarrow G$ such that $q^* \chi$ is the differential of a character of $\tilde{G}$. Then one can consider the monodromic derived category $D_{\xi,\theta}^b(Y)$, which is the ξ -isotypic component of $\tilde{G}$ -equivariant derived category. For $G = \mathrm{GL}_n$, take $\tilde{G} = \mathrm{SL}_n \times \mathbb{C}^\times$ and $\eta(g, t) = t^d$ for $(g, t) \in \mathrm{SL}_n \times \mathbb{C}^\times$ and $d \in \mathbb{Z}$ giving $\xi = \det^{d/n}$. The horocycle correspondence (5), induction as well the functors CH and HC are adaptable in this setting.

2.5. Singular supports. Fix a nondegenerate invariant bilinear form on $\mathfrak{g}$ and use it to identify $\mathfrak{g} \cong \mathfrak{g}^*$. Use $\cdot$ to denote the adjoint action. Consider the moment map $\mu_G : T^*G \cong G \times \mathfrak{g} \rightarrow \mathfrak{g}$, $(g, x) \mapsto g \cdot x - x$. Define the derived commuting scheme $\mathfrak{C}$ to be the derived zero fiber of $\mu_G : T^*G \cong G \times \mathfrak{g} \rightarrow \mathfrak{g}$. Its classical truncation is $t_0\mathfrak{C} = \{(g, x) \in G \times \mathfrak{g} \mid g \cdot x = x\}$. The $G \times \mathbb{C}^\times$ action on $G \times \mathfrak{g}$:

$$(g, z)(h, x) = (ghg^{-1}, zg \cdot x), \quad g, h \in G, \quad z \in \mathbb{C}^\times, \quad x \in \mathfrak{g}$$

restricts to a $G \times \mathbb{C}^\times$ action on $\mathfrak{C}$.

Moreover,

$$T^*Y \cong \{(x, y, \alpha, \beta) \in Y \times \mathfrak{g} \times \mathfrak{g} \mid x^{-1} \cdot \alpha, y^{-1} \cdot \beta \in \mathfrak{b}, x^{-1} \cdot \alpha \equiv y^{-1} \cdot \beta \bmod \mathfrak{n}\}.$$

Consider the moment map:

$$\mu_Y : T^*Y \rightarrow \mathfrak{g} \times \mathfrak{t}, (x, y, \alpha, \beta) \mapsto (\alpha - \beta, y^{-1} \cdot \beta \bmod \mathfrak{n}).$$

Let $\tilde{\mathcal{N}} := G \times^B \mathfrak{n}$. The derived zero fiber $\tilde{\mathcal{Z}}^R$ of μ_Y is isomorphic to a T -torsor over $\mathcal{Z}^R := \tilde{\mathcal{N}} \times_{\mathfrak{g}}^R \tilde{\mathcal{N}}$, pullback from $Y \rightarrow \mathcal{B}^2$. The $G \times \mathbb{C}^\times$ -action on $\tilde{\mathcal{N}}$:

$$(g, z)(\mathfrak{b}, x) = (g \cdot \mathfrak{b}, zg \cdot x)$$

induces a $G \times \mathbb{C}^\times$ -action on $\mathcal{Z}^R$.

For a derived scheme X with an action of an affine algebraic group H , let $\text{Coh}_H(X)$ denote the full subcategory of the quasi-coherent derived category consisting of complexes with bounded coherent cohomology. Let $\text{Coh}_H(X)^\heartsuit$ denote the heart of $\text{Coh}_H(X)$ under the usual t-structure.

Let $\text{Coh}_{G \times \mathbb{C}^\times, \mathcal{N}}(\mathfrak{C})$ denote the full subcategory of $\text{Coh}_{G \times \mathbb{C}^\times, \mathcal{N}}(\mathfrak{C})$ whose objects have set-theoretic support contained in $G \times \mathcal{N}$.

Lemma 2.6. *Taking associated graded with respect to the Hodge filtration canonically defines functors*

$$\begin{aligned} \text{gr}_{\mathcal{CS}} : \mathcal{CS}_\theta &\rightarrow \text{Coh}_{\mathcal{N}, G \times \mathbb{C}^\times}(\mathfrak{C}), \\ \tilde{\text{gr}}_{\mathcal{H}} : \mathcal{H}_\theta &\rightarrow \text{Coh}_{G \times \mathbb{C}^\times}(\tilde{\mathcal{Z}}^R). \end{aligned}$$

The second functor naturally retains $\tilde{T}$ -equivariance, with $\ker \alpha$ acting by θ .

Proof. Each assertion stems from the compatibility between the Hodge filtration on a module and the filtration on the sheaf of algebras by the order of differential operator. In particular, in the first claim, $\text{gr}^F(N)$ naturally forms a module over $\text{gr}^F \mathcal{D}_G = \mathcal{O}_{T^*G}$.

We first check that the ideal generated by the image of the comoment map μ_G^* annihilates $\text{gr}^F(N)$ scheme-theoretically. For $\xi \in \mathfrak{g}$ write ξ_G for the corresponding vector field and L_ξ for the derivative of the $(G, \cdot)$ -action on N . The strong G -equivariance of N implies that $\xi_G = L_\xi$ as operators on N . Since N underlies a G -equivariant mixed Hodge module, its Hodge filtration is G -equivariant, so L_ξ preserves each $F_k N$. Hence ξ_G maps $F_k N$ into $F_k N$, and therefore its symbol in $\text{gr}_1^F \mathcal{D}_G$ acts by 0 on $\text{gr}^F(N)$. That symbol is exactly $\mu_G^*(\xi)$. Thus $\mu_G^*(\mathfrak{g})\text{gr}^F(N) = 0$ and $\text{gr}^F(N)$ is a module over the classical zero fiber $t_0\mathfrak{C} = \mu_G^{-1}(0)$. Moreover, it is well-known (e.g in [Gi]) that $\text{gr}^F(N)$ has set-theoretic support in $G \times \mathcal{N}$.

It remains to promote this to a module over the derived zero fiber $\mathfrak{C}$. But the canonical truncation

$$\mathcal{O}_{\mathfrak{C}} \longrightarrow \tau^{\geq 0} \mathcal{O}_{\mathfrak{C}} = H^0(\mathcal{O}_{\mathfrak{C}}) = \mathcal{O}_{t_0\mathfrak{C}}$$

is a map of commutative dg algebras, and we can thus pushforward $\mathrm{gr}^F(N)$ to $\mathfrak{C}$. The $\mathbb{C}^\times$ -equivariance comes from the natural grading on gr^F .

The proof of the second statement proceeds analogously by replacing μ_G with μ_Y , where the T -equivariance arises because the monodromy of a (T, θ) -monodromic module becomes trivial upon taking the associated graded. $\square$

Moreover, equivariant descent along the T -bundle $\tilde{\mathcal{Z}}^R \rightarrow \mathcal{Z}^R$ gives a t-exact equivalence

$$\mathrm{Desc}_T : \mathrm{Coh}_{G \times T \times \mathbb{C}^\times}(\tilde{\mathcal{Z}}^R) \xrightarrow{\sim} \mathrm{Coh}_{G \times \mathbb{C}^\times}(\mathcal{Z}^R).$$

For every $M \in \mathcal{H}_\theta$, $\tilde{\mathrm{gr}}_{\mathcal{H}}(M)$ is $\tilde{T}$ -equivariant. Let $\tilde{\lambda} \in \mathrm{Lie}(\tilde{T})^*$ be the integral lift of λ . Then the kernel of the finite cover $\mu : \tilde{T} \rightarrow T$ acts on $\tilde{\mathrm{gr}}_{\mathcal{H}}(M)$ by $\tilde{\lambda}$. Thus $\tilde{\mathrm{gr}}_{\mathcal{H}}(M) \otimes \mathbb{C}_{-\tilde{\lambda}}$ is T -equivariant. We define

$$\mathrm{gr}_{\mathcal{H}}(M) := \mathrm{Desc}_T(\tilde{\mathrm{gr}}_{\mathcal{H}}(M) \otimes \mathbb{C}_{-\tilde{\lambda}}).$$

Write $D_\lambda(F) := \mathrm{Desc}_T(F \otimes \mathbb{C}_{-\tilde{\lambda}})$.

2.6. The coherent character functor $\widetilde{\mathrm{CH}}$. Let

$$\mathfrak{X} := \{(g, h, x) \in G \times \mathcal{B} \times \mathfrak{g} \mid g^{-1} \cdot x \in h \cdot \mathfrak{n}\}.$$

Define $\mathfrak{Y}$ to be the derived fiber product of the following diagram.

$$\begin{array}{ccc} T^*\mathcal{B}^2 & \longleftarrow & \mathfrak{X} \\ \uparrow & & \uparrow \\ \mathcal{Z}^R & \xleftarrow{r} & \mathfrak{Y} \end{array}$$

Since $\mathcal{Z}^R$ is the derived zero fiber of $\mu_{\mathcal{B}^2}$, $\mathfrak{Y}$ is the derived zero fiber of the composition $\mathfrak{X} \rightarrow T^*\mathcal{B}^2 \xrightarrow{\mu_{\mathcal{B}^2}} \mathfrak{g}$. Explicitly, the image of a point $(g, h, x) \in \mathfrak{X}$ in $T^*_{(h, gh)}\mathcal{B}^2$ is the pair $(-g^{-1} \cdot x, x)$, so this composition is $(g, h, x) \mapsto x - g^{-1} \cdot x$; composing with the automorphism $y \mapsto -g \cdot y$ of the trivial bundle $\mathfrak{X} \times \mathfrak{g}$ we may equally write it as $(g, h, x) \mapsto g \cdot x - x$. In summary, $\mathfrak{Y}$ is the derived zero fiber of the map $\mathfrak{X} \rightarrow \mathfrak{g}$, $(g, h, x) \mapsto g \cdot x - x$.

Let $s : \mathfrak{Y} \rightarrow \mathfrak{C}$, $(g, h, x) \mapsto (g, x)$. For a map $V \rightarrow \mathcal{B}$ from a variety V , let $\omega_{\mathcal{B}}^V$ denote the pullback of the canonical bundle of $\mathcal{B}$ to V . Write $\mathcal{AH} := \mathrm{Coh}_{G \times \mathbb{C}^\times}(\mathcal{Z}^R)$. Let $b = \dim(\mathcal{B})$.

Definition 2.7. Define the functor

$$\widetilde{\mathrm{CH}}(-) := R s_* (L r^*(-) \otimes \omega_{\mathcal{B}}^{\mathfrak{Y}} \langle -b \rangle) : \mathcal{AH} \longrightarrow \mathrm{Coh}_{\mathcal{N}, G \times \mathbb{C}^\times}(\mathfrak{C}).$$

Theorem 2.8. *The diagram*

$$(6) \quad \begin{array}{ccc} \mathcal{H}_\theta & \xrightarrow{\mathrm{CH}} & \mathcal{CS}_\theta \\ \downarrow \mathrm{gr}_{\mathcal{H}} & & \downarrow \mathrm{gr}_{\mathcal{CS}} \\ \mathcal{AH} & \xrightarrow{\widetilde{\mathrm{CH}}} & \mathrm{Coh}_{\mathcal{N}, G \times \mathbb{C}^\times}(\mathfrak{C}) \end{array}$$

commutes.

Proof. Denote $G \times \mathcal{B}$ by X . Combining the two Lagrangian correspondences (1) for the maps a, p from (5), we obtain:

$$(7) \quad \begin{array}{ccccc} T^*Y & \xleftarrow{\varpi_a} & T^*Y \times_Y X & \xrightarrow{\rho_a} & T^*X \\ & \nwarrow \rho & \uparrow & & \uparrow \rho_p \\ & & \tilde{\mathfrak{X}} & \xrightarrow{\quad} & T^*G \times_G X \\ & & & \searrow \varpi & \downarrow \varpi_p \\ & & & & T^*G \end{array}$$

Define $\tilde{\mathfrak{X}}$ to be the fiber product in the square in diagram (7). We first show that

$$(8) \quad \tilde{\mathfrak{X}} \cong \{(g, \mathfrak{b}', x) \in G \times \mathcal{B} \times \mathfrak{g} \mid g^{-1} \cdot x \in \mathfrak{b}'\}.$$

By G -equivariance it suffices to compute the fiber of $\tilde{\mathfrak{X}}$ at $(g, \mathfrak{b}) \in G \times \mathcal{B}$. Right-translation trivializations give $T_{(g, \mathfrak{b})}^*X \cong \mathfrak{g} \times \mathfrak{n}$ and

$$T_{a(g, \mathfrak{b})}Y \cong (\mathfrak{g}/\mathfrak{n} \times \mathfrak{g}/(g \cdot \mathfrak{n}))/\mathfrak{t}_g, \quad \mathfrak{t}_g := \{(\tau, g \cdot \tau) \mid \tau \in \mathfrak{t}\}.$$

For a lift $\hat{\eta} \in \mathfrak{g}$ of $\eta \in \mathfrak{g}/\mathfrak{b}$, the differentials are

$$\begin{aligned} da : \mathfrak{g} \times \mathfrak{g}/\mathfrak{b} &\rightarrow (\mathfrak{g}/\mathfrak{n} \times \mathfrak{g}/(g \cdot \mathfrak{n}))/\mathfrak{t}_g, & (\xi, \eta) &\mapsto \overline{(\hat{\eta}, \xi + g \cdot \hat{\eta})} \\ dp : \mathfrak{g} \times \mathfrak{g}/\mathfrak{b} &\rightarrow \mathfrak{g}, & (\xi, \eta) &\mapsto \xi. \end{aligned}$$

A direct check gives

$$\text{Ker}(da) = (g \cdot \mathfrak{n}) \times \{0\}, \quad \text{Ker}(dp) = \{0\} \times \mathfrak{g}/\mathfrak{b}.$$

Since the dimension of $\text{Ker}(da)$ is constant, we see that a is smooth, so the images of the two codifferentials are the annihilators of these kernels. Hence

$$(9) \quad \tilde{\mathfrak{X}}_{(g, \mathfrak{b})} = \text{Im}(d^*a) \cap \text{Im}(d^*p) = (\text{Ker}(da) + \text{Ker}(dp))^\perp = (g \cdot \mathfrak{b}) \times \{0\},$$

and (8) follows. Moreover, since $\text{Ker}(da) \cap \text{Ker}(dp) = \{0\}$, the second term in (9) is also the derived intersection.

Let $\omega_{\mathcal{B}}^{\tilde{\mathfrak{X}}}$ denote the pullback of the canonical bundle of $\mathcal{B}$ to $\tilde{\mathfrak{X}}$. Now using Theorem 2.1, Lemma 2.2, and base change, we have that

$$(10) \quad \text{gr}(\text{CH}(M, F_\bullet)) = R\varpi_*(L\rho^*(\text{gr}^F(M)) \otimes \omega_{\mathcal{B}}^{\tilde{\mathfrak{X}}} \langle -b \rangle).$$

Take $\tilde{\mathfrak{Y}} := \tilde{\mathfrak{X}} \times_{T^*Y} \tilde{\mathfrak{Z}}^R$. Note that $\tilde{\mathfrak{Y}}$ is the derived subscheme of $\tilde{\mathfrak{X}}$ cut off by the conditions $g \cdot x = x$ and $g^{-1} \cdot x = 0$ in $h \cdot \mathfrak{b}/h \cdot \mathfrak{n}$. Therefore, $\tilde{\mathfrak{Y}} \cong \mathfrak{Y}$. Let $\tilde{r} : \mathfrak{Y} \cong \tilde{\mathfrak{Y}} \rightarrow \tilde{\mathfrak{Z}}^R$ be the composition map.

Write $i : \mathfrak{C} \rightarrow T^*G$ for the natural inclusion. By base change, the right hand side of (10) equals

$$Ri_*Rs_*(L\tilde{r}^*(\tilde{\text{gr}}_{\mathcal{H}}(M)) \otimes \omega_{\mathcal{B}}^{\mathfrak{Y}}).$$

Now in $\text{Coh}_{\mathcal{N}, G \times \mathbb{C}^\times}(\mathfrak{C})$ we have,

$$Rs_*(L\tilde{r}^*(\tilde{\text{gr}}_{\mathcal{H}}(M)) \otimes \omega_{\mathcal{B}}^{\mathfrak{Y}}) = Rs_*(Lr^*(\text{gr}_{\mathcal{H}}(M)) \otimes \omega_{\mathcal{B}}^{\mathfrak{Y}})$$

and the theorem follows. $\square$

3. ASSOCIATED GRADED OF (CO)STANDARD OBJECTS

We compute the image of standard and costandard objects in $\mathcal{H}_\theta$ introduced in (3) and (4) under the map $\mathrm{gr}_{\mathcal{H}}$ from the diagram (6). Note that $\nabla_{w,\theta}^e$ and $\Delta_{w,\theta}^e$ are obtained by taking the $\dagger$ and $!$ extension to the Bott-Samuelson resolution $\widetilde{BS}_w$ of the local system (2), and then pushing forward to $\widetilde{\mathcal{B}}$. Given the result of Laumon (Theorem 2.1), one may carry out the computation on $\widetilde{BS}_w$ where the extension is across a simple normal crossing (SNC) divisor. In this setting, Saito's theory describes the Hodge filtration explicitly in terms of Deligne's canonical lattice.

Here is our motivation for this computation. As we mentioned in the introduction the main computation of this paper is that of $\widetilde{\mathrm{CH}}(\omega^a)$, a $\mathrm{GL}_r \times \mathbb{C}^\times$ -equivariant coherent sheaf on the commuting stack $\mathfrak{C}_r$. The direct computation seems to be very hard if at all possible. Instead we use Theorem 2.8. It turns out that the object ω^a (we will work with a version of such objects for SL_r) arises as the Hodge associated graded of $\nabla_{w,\theta}$ – that in the case of interest coincides with $\Delta_{w,\theta}$ – for suitable w, θ . The main results of these section, Theorem 3.4 and its special case, Corollary 3.6 explain how to compute Hodge associated graded sheaves of these objects so that in the case of interest we indeed recover ω^a .

3.1. Steinberg variety. Let $\mathcal{Z}_{\mathfrak{g}} := \widetilde{\mathfrak{g}} \times_{\mathfrak{g}}^R \widetilde{\mathfrak{g}}$, which coincides with the nonderived fiber product. Take $\mathcal{Z} := t_0(\mathcal{Z}^R) = \widetilde{\mathcal{N}} \times_{\mathcal{N}} \widetilde{\mathcal{N}}$.

In the regular semisimple part $\widetilde{\mathfrak{g}}^{rs} \times_{\mathfrak{g}^{rs}} \widetilde{\mathfrak{g}}^{rs}$, let $\Lambda_{\mathfrak{g},w}$ be the graph of w such that its image in $\mathcal{B}^2$ is the orbit of (B, wB) . We write

$$\Lambda_w := \overline{\Lambda_{\mathfrak{g},w}} \cap \mathcal{Z}$$

for the scheme-theoretic intersection inside $\mathcal{Z}_{\mathfrak{g}}$.

Define

$$\mathcal{Z}^e := \{(h, x) \in \mathcal{B} \times \mathfrak{b} \mid x \in h \cdot [\mathfrak{b}, \mathfrak{b}]\}.$$

Conjugation induces an isomorphism $f : G \times^B \mathcal{Z}^e \xrightarrow{\sim} \mathcal{Z}$, and

$$\mathcal{Z}^e = \bigsqcup_{w \in W} T_{\mathcal{B}_w}^* \mathcal{B}.$$

Similar to $\mathcal{Z}^e$, take

$$\mathcal{Z}_{\mathfrak{g}}^e := \{(\mathfrak{b}', x) \in \mathcal{B} \times \mathfrak{b}, x \in \mathfrak{b}'\}.$$

There is an isomorphism $\mathcal{Z}_{\mathfrak{g}} \cong G \times^B \mathcal{Z}_{\mathfrak{g}}^e$. The Borel subalgebras that contain a given regular semisimple element are in bijection with W . Therefore, writing $\Lambda_{\mathfrak{g},w}^e = \{(\mathfrak{b}', x) \in \mathcal{B}_w \times \mathfrak{b}^{rs}, x \in \mathfrak{b}'\}$, we have

$$\mathcal{Z}_{\mathfrak{g}}^e := \{(\mathfrak{b}', x) \in \mathcal{B} \times \mathfrak{b}^{rs}, x \in \mathfrak{b}'\} = \sqcup_w \Lambda_{\mathfrak{g},w}^e.$$

We may rewrite the graph $\Lambda_{\mathfrak{g},w} \subset \mathcal{Z}_{\mathfrak{g}}$ by $\Lambda_{\mathfrak{g},w} \cong G \times^B \Lambda_{\mathfrak{g},w}^e$. As a result,

$$\Lambda_w \cong G \times^B \Lambda_w^e, \text{ where } \Lambda_w^e := (\overline{\Lambda_{\mathfrak{g},w}^e} \cap \mathcal{Z}^e).$$

Let $\mathcal{O}_X(\alpha)$ be the pullback of the line bundle on $\mathcal{B}$ corresponding to the weight α , i.e., $\mathrm{ind}_{\mathcal{B}}^G \mathbb{C}_{-\alpha}$. Under this convention, dominant weights induce ample line bundles on $\mathcal{B}$. For $\mu, \nu \in X^*(T)$, there is a line bundle $\mathcal{O}_{\mathcal{Z}}(\mu, \nu)$ on $\mathcal{Z}$ restricted from $\mathcal{O}_{\widetilde{\mathcal{N}}}(\mu) \boxtimes \mathcal{O}_{\widetilde{\mathcal{N}}}(\nu)$. Let ρ be the half sum of positive roots. We note that even when ρ is not in $X^*(T)$, $\mathcal{O}_{\mathcal{Z}}(\rho, \rho)$ is still well-defined.

3.2. Bott-Samelson resolution. Throughout this subsection $w \in W$ is fixed together with a reduced expression $w = s_{k_1} \cdots s_{k_\ell}$. For a simple reflection s , write P_s for the associated minimal parabolic. Let

$$\widetilde{BS}_w := P_{s_{k_1}} \times^B \cdots \times^B P_{s_{k_\ell}} / U, \quad BS_w := P_{s_{k_1}} \times^B \cdots \times^B P_{s_{k_\ell}} / B.$$

Let $\tilde{p} : \widetilde{BS}_w \rightarrow \widetilde{\mathcal{B}}$ and $p : BS_w \rightarrow \mathcal{B}$ be the multiplication maps. We identify BS_w with the variety of sequences $(\mathfrak{b}_0, \dots, \mathfrak{b}_\ell)$ of Borel subalgebras with $\mathfrak{b}_0 = \mathfrak{b}$ and $\mathfrak{b}_{j-1}, \mathfrak{b}_j$ in relative position $\leq s_{k_j}$, and set

$$E_j := \{\mathfrak{b}_{j-1} = \mathfrak{b}_j\} \subset BS_w, \quad D_j := E_j \times_{\mathcal{B}} \widetilde{\mathcal{B}} \subset \widetilde{BS}_w.$$

Write $r_w : \widetilde{BS}_w \rightarrow BS_w$ for the T -torsor, canonically identified with the pullback of $\widetilde{\mathcal{B}} \rightarrow \mathcal{B}$:

$$\widetilde{BS}_w \xrightarrow{\sim} BS_w \times_{\mathcal{B}} \widetilde{\mathcal{B}}, \quad [p_1, \dots, p_\ell]_U \mapsto ([p_1, \dots, p_\ell]_B, p_1 \cdots p_\ell U).$$

The divisors $E_1, \dots, E_\ell$ are smooth and meet transversally; their complement is isomorphic to the open cell $\mathcal{B}_w$. Writing $E := \sum_j E_j$, we also have that

$$D := \widetilde{BS}_w \setminus \widetilde{\mathcal{B}}_w = \bigcup_{j=1}^{\ell} D_j = r_w^* E,$$

which is an SNC divisor.

Note that $\widetilde{\mathcal{B}}_w$ is a T -bundle over the affine space $\mathcal{B}_w$ so $\pi_1(\widetilde{\mathcal{B}}_w) \xrightarrow{\sim} \pi_1(T)$. On the other hand, $\pi_1(T)$ is naturally identified with the cocharacter lattice $X_*(T)$. For $1 \leq j \leq \ell$, put

$$(11) \quad \gamma_j := s_{k_\ell} s_{k_{\ell-1}} \cdots s_{k_{j+1}} (\alpha_{k_j}^\vee).$$

Lemma 3.1. *A positive meridian around D_j represents the class γ_j in $\pi_1(\widetilde{\mathcal{B}}_w)$. By a positive meridian, we mean a counterclockwise loop in a transverse complex disk at a smooth point of D_j based in the complement.*

Proof. Let $\beta_j \in X_*(T) \cong \pi_1(\widetilde{\mathcal{B}}_w)$ be the class of a positive meridian about D_j . We first show in the base case when $w = s_{j_k}$ that $\beta_j = \alpha_{k_j}^\vee$.

Let L be the Levi subgroup of $P_{s_{j_k}}$. Then $P_{s_{j_k}}/U \cong L/L \cap U$. Therefore it suffices to study the case when G is of semisimple rank 1, i.e, SL_2 , PGL_2 or GL_2 . We know that G/U is

- the $\mathbb{C}^\times$ -bundle on $\mathbb{P}^1$ associated to $\mathcal{O}(-1)$ for SL_2 .
- the $\mathbb{C}^\times$ -bundle on $\mathbb{P}^1$ associated to $\mathcal{O}(-2)$ for PGL_2 .
- the $(\mathbb{C}^\times)^2$ -bundle on $\mathbb{P}^1$ associated to $\mathcal{O}(-1) \oplus \mathcal{O}(1)$ for GL_2 .

Let $s = (12)$ be the simple reflection. According to Lemma 3.2 below, a positive meridian around the complement of $\widetilde{\mathcal{B}}_s$ corresponds to 1, 2 or $(1, -1)$ in $X_*(T)$ respectively, which is exactly the coroot $\alpha_s^\vee$ in each case. Therefore the base case is proved.

In the iterated Bott-Samelson construction, the factors to its right transport this cocharacter by $s_{k_\ell} \cdots s_{k_{j+1}}$. Therefore

$$\beta_j = s_{k_\ell} \cdots s_{k_{j+1}} (\alpha_{k_j}^\vee) = \gamma_j.$$

□

Take $k \in \mathbb{Z}$. Let X be the total space of $\mathcal{O}_{\mathbb{P}^1}(k)$ minus the zero section and the fiber at ∞ . Then the isomorphism $X \cong \mathbb{C} \times \mathbb{C}^\times$ induces the identification $\pi_1(X) \cong \pi_1(\mathbb{C}^\times) \cong \mathbb{Z}$.

Lemma 3.2. *A positive meridian around the fiber at ∞ in X corresponds to $-k \in \pi_1(X)$.*

Proof. Let (z, t_0) and (w, t_∞) be the coordinates of the affine chart of $\text{Tot}(\mathcal{O}_{\mathbb{P}^1}(k))$ at 0 and ∞ respectively. Then $w = 1/z$ and $t_\infty = w^k t_0$. Take a positive meridian $w(\theta) = \epsilon e^{i\theta}$ around the fiber at ∞ for $0 < \epsilon \ll 1$ and $t_\infty \equiv s$ for some constant s . Then in the chart around 0, $t_0 = \frac{s}{\epsilon} e^{-ki\theta}$, which represents the class $-k \in \pi_1(X)$. $\square$

3.3. Deligne lattices. Let $\mathbf{L}_{w,\theta}$, resp. $\mathbf{L}_{w,\theta}^!$, denote the $\dagger$ -, resp. $!$ -extension of $\mathbb{L}_{w,\theta}$ from $\widetilde{\mathcal{B}}_w$ to $\widetilde{BS}_w$, and let $\mathbf{L}_{w,\theta}^{\geq -1} \subset \mathbf{L}_{w,\theta}$, resp. $\mathbf{L}_{w,\theta}^{!> -1} \subset \mathbf{L}_{w,\theta}^!$, be Deligne's canonical lattice [Sai2, (3.10)], i.e. the lattice whose residue eigenvalues along each component of D lie in $[-1, 0)$, resp. $(-1, 0]$.

Fix a rational logarithm $\lambda \in X^*(T) \otimes \mathbb{Q}$ of θ and set

$$c_j := \langle \lambda, \gamma_j \rangle, \quad n_j := \lfloor c_j \rfloor + 1, \quad m_j := \lceil c_j \rceil, \quad f_j := c_j - n_j, \quad e_j := c_j - m_j.$$

Thus $f_j \in [-1, 0)$ and $e_j \in (-1, 0]$. On BS_w , set

$$P_\lambda := \sum_{j=1}^{\ell} \lfloor c_j \rfloor E_j, \quad E_\theta := \sum_{j: \theta(\gamma_j)=1} E_j,$$

and write $F_\lambda := r_w^* P_\lambda$ and $D_\theta := r_w^* E_\theta$.

Lemma 3.3. *We have isomorphisms of line bundles*

$$(12) \quad \mathbf{L}_{w,\theta}^{\geq -1} \cong \mathcal{O}(D + F_\lambda), \quad \mathbf{L}_{w,\theta}^{!> -1} \cong \mathcal{O}(D + F_\lambda - D_\theta) \cong \mathbf{L}_{w,\theta}^{\geq -1}(-D_\theta).$$

Proof. Let $\mathcal{M}_{w,\theta} := \mathbf{L}_{w,\theta}(*D) \cong \mathbf{L}_{w,\theta}^!(*D)$ be the common meromorphic extension of $\mathbb{L}_{w,\theta}$ to $\widetilde{BS}_w$. Here $\mathbf{L}_{w,\theta}(*D)$, resp. $\mathbf{L}_{w,\theta}^!(*D)$, denotes the sheaf of rational sections that are regular away from D but with possible poles along D . Consider the product presentation $U_w \times T \xrightarrow{\sim} \widetilde{\mathcal{B}}_w$, $(u, t) \mapsto uwtU$ and let $t : \widetilde{\mathcal{B}}_w \rightarrow T$ denote the projection.

Recall $\mathbb{L}_{w,\theta}$ defined by (2). Write

$$\mathbb{L}_{w,\theta} \cong \mathcal{O}_{U_w} \boxtimes \mathcal{K}_\lambda, \quad \text{where } \mathcal{K}_\lambda := \mathcal{D}_T / \sum_{\xi \in \mathfrak{t}} \mathcal{D}_T (\partial_\xi - \langle \lambda, \xi \rangle).$$

Let ψ_0 be the section of $\mathbb{L}_{w,\theta}$ corresponding to $1 \boxtimes 1$. It determines a unique rational section ψ of $\mathcal{M}_{w,\theta}$ such that $\mathcal{M}_{w,\theta} = \mathcal{O}_{\widetilde{BS}_w}(*D)\psi$.

Let $V \subset \widetilde{BS}_w$ be a SNC chart with $V \cong \mathbb{A}^{\ell(w)} \times T$. Let J_V index the boundary components meeting V . Let $q_{V,j}$ be the defining equations of $D_j \cap V$ pulled back from $BS_w \cap r_w(V)$, invariant under the T -action.

By Lemma 3.1, we have

$$\text{ord}_{D_j} \chi(t) = \langle \chi, \gamma_j \rangle \quad \text{for } \chi \in X^*(T).$$

Let $\tau_V : V \rightarrow T$ be the projection. Then there is a regular map $a_V : V \rightarrow T$ such that, on $V \cap \widetilde{\mathcal{B}}_w$,

$$(13) \quad t = a_V \left(\prod_{j \in J_V} \gamma_j(q_{V,j}) \right) \tau_V.$$

For $j \in J_V$, put

$$\eta_{V,j} := (d \log a_V) (q_{V,j} \partial_{q_{V,j}}) \in \mathcal{O}(V) \otimes \mathfrak{t}.$$

Applying $q_{V,j}\partial_{q_{V,j}}$ to (13) and using the defining equations of ψ , we obtain

$$(14) \quad q_{V,j}\partial_{q_{V,j}}\psi = (c_j + \langle \lambda, \eta_{V,j} \rangle) \psi.$$

Since $\eta_{V,j} \in q_{V,j}\mathcal{O}(V) \otimes \mathfrak{t}$, the function $\langle \lambda, \eta_{V,j} \rangle$ vanishes on D_j , we have $\text{Res}_{D_j}(\psi) = c_j$.

Now set

$$\phi_V := \left(\prod_{j \in J_V} q_{V,j}^{-n_j} \right) \psi, \quad \phi_V^! := \left(\prod_{j \in J_V} q_{V,j}^{-m_j} \right) \psi.$$

Their residues along D_j are, respectively,

$$c_j - n_j = f_j \in [-1, 0), \quad c_j - m_j = e_j \in (-1, 0].$$

It follows that

$$(15) \quad \mathbf{L}_{w,\theta}^{\geq -1}(V) = \mathcal{O}(V)\phi_V, \quad \mathbf{L}_{w,\theta}^{!> -1}(V) = \mathcal{O}(V)\phi_V^!$$

which are understood via injective maps into $\mathcal{M}_{w,\theta}$. At the intersection of two charts V and V' , ϕ_V and $\phi_{V'}$ differ by the unit

$$\alpha_{V,V'} := \prod_{j \in J_V} q_{V,j}^{n_j} \prod_{j \in J_{V'}} q_{V',j}^{-n_j}$$

Since $(\alpha_{V,V'})$ are the transition functions of the line bundle $\mathcal{O}(\sum_j n_j D_j)$, we obtain that

$$\mathbf{L}_{w,\theta}^{\geq -1} \cong \mathcal{O}_{\widetilde{BS}_w} \left(\sum_j n_j D_j \right).$$

Similarly, $\mathbf{L}_{w,\theta}^{!> -1} \cong \mathcal{O}_{\widetilde{BS}_w} \left(\sum_j m_j D_j \right)$.

Finally,

$$n_j = \lfloor c_j \rfloor + 1, \quad m_j = \begin{cases} \lfloor c_j \rfloor + 1, & c_j \notin \mathbb{Z}, \\ \lfloor c_j \rfloor, & c_j \in \mathbb{Z}, \end{cases}$$

and $c_j \in \mathbb{Z}$ if and only if $\theta(\gamma_j) = 1$. Therefore

$$\sum_j n_j D_j = D + F_\lambda, \quad \sum_j m_j D_j = D + F_\lambda - D_\theta.$$

Consequently,

$$\mathbf{L}_{w,\theta}^{\geq -1} \cong \mathcal{O}(D + F_\lambda), \quad \mathbf{L}_{w,\theta}^{!> -1} \cong \mathcal{O}(D + F_\lambda - D_\theta) \cong \mathbf{L}_{w,\theta}^{\geq -1}(-D_\theta),$$

as desired. $\square$

3.4. gr of (co)standards. Let $\widetilde{\Sigma}_w \subset T^*\widetilde{BS}_w$ be the closed subscheme of covectors that annihilate all logarithmic vector fields in $T_{\widetilde{BS}_w}(-\log D)$. With respect to the trivialization $V \simeq V_0 \times T$, choose boundary coordinates $q_{V,j}$, remaining coordinates $x_{V,a}$ in the V_0 -direction, and torus coordinates $z_{V,b}$. If $\xi_{V,j}$, $\eta_{V,a}$, and $\zeta_{V,b}$ denote the principal symbols of $\partial_{q_{V,j}}$, $\partial_{x_{V,a}}$, and $z_{V,b}\partial_{z_{V,b}}$, then

$$\widetilde{\Sigma}_w \cap T^*V = Z((q_{V,j}\xi_{V,j})_{j \in J_V}, (\eta_{V,a})_a, (\zeta_{V,b})_{b=1}^r)$$

where $Z(-)$ on the right hand side indicates vanishing locus of the listed functions. Its underlying set is the union of the conormal bundles to the boundary strata. Let $\Sigma_w \subset T^*BS_w$ be its quotient by T , and write $\widetilde{\iota}_w$ and ι_w for the respective closed embeddings.

For the map $p : BS_w \rightarrow \mathcal{B}$, put

$$(16) \quad A_w^e := BS_w \times_{p, \mathcal{B}} T^* \mathcal{B}, \quad T^* BS_w \xleftarrow{\rho_p} A_w^e \xrightarrow{\varpi_p} T^* \mathcal{B},$$

where ρ_p is the codifferential and ϖ_p is the projection. Define

$$\begin{aligned} (\mathbf{K}_{w, \lambda}^\nabla)^e &:= (\varpi_p)_* (\rho_p^* ((\iota_w)_* \mathcal{O}_{\Sigma_w}) \otimes \mathcal{O}(P_\lambda)) , \\ (\mathbf{K}_{w, \lambda}^\Delta)^e &:= (\varpi_p)_* (\rho_p^* ((\iota_w)_* \mathcal{O}_{\Sigma_w}) \otimes \mathcal{O}(P_\lambda - E_\theta)) . \end{aligned}$$

Here and blow divisor line bundles are pulled back to A_w^e . Let $\mathbf{K}_{w, \lambda}^\nabla := \text{ind}_B^G (\mathbf{K}_{w, \lambda}^\nabla)^e$ and $\mathbf{K}_{w, \lambda}^\Delta := \text{ind}_B^G (\mathbf{K}_{w, \lambda}^\Delta)^e$.

Theorem 3.4. *Suppose $w \in W_\theta$. Then under the definitions (3) and (4), we have that, in $\text{Coh}_{G \times \mathbb{C}^\times}(\mathcal{Z}^R)$*

$$(17) \quad \text{gr}_{\mathcal{H}}(\nabla_{w, \theta}) \cong \mathbf{K}_{w, \lambda}^\nabla \otimes \mathcal{O}(w\lambda - \lambda + \rho, \rho) \langle b - \ell(w) \rangle ,$$

$$(18) \quad \text{gr}_{\mathcal{H}}(\Delta_{w, \theta}) \cong \mathbf{K}_{w, \lambda}^\Delta \otimes \mathcal{O}(w\lambda - \lambda + \rho, \rho) \langle b - \ell(w) \rangle .$$

Proof. By Proposition 2.3, it suffices to compute $\text{gr} \nabla_{w, \theta}^e$ and $\text{gr} \Delta_{w, \theta}^e$ as $\text{Ad}(B)$ -equivariant coherent sheaves. We first compute $\text{gr}(\mathbf{L}_{w, \theta})$ and $\text{gr}(\mathbf{L}_{w, \theta}^!)$ on $\widehat{BS}_w$ and then apply Theorem 2.1.

Quotient description. We first compute $\text{gr} \nabla_{w, \theta}^e$ and $\text{gr} \Delta_{w, \theta}^e$ without the $\text{Ad}(B)$ -equivariance. On the SNC chart $V \simeq V_0 \times T$ with basis $q_{V, j}, x_{V, a}, z_{V, b}$ as above, write $z_{V, b} = \chi_b \circ \tau_V$ for a fixed basis $\chi_1, \dots, \chi_r$ of $X^*(T)$. Let $\xi_1, \dots, \xi_r$ be the dual basis of $X_*(T)$, and set $\lambda_b := \langle \lambda, \xi_b \rangle$.

Using (13), set

$$A_{V, j} := \langle \lambda, (d \log a_V)(q_{V, j} \partial_{q_{V, j}}) \rangle , \quad A_{V, a} := \langle \lambda, (d \log a_V)(\partial_{x_{V, a}}) \rangle .$$

The right T -equivariance of t and τ_V implies that a_V is constant along the T -direction. Hence a_V contributes no term in the $z_{V, b}$ -directions.

With respect to the local generators ϕ_V and $\phi_V^!$ of (15), we have

$$(19)$$

$$\mathbf{L}_{w, \theta}|_V \cong \mathcal{D}_V / \left(\sum_{j \in J_V} \mathcal{D}_V (q_{V, j} \partial_{q_{V, j}} - f_j - A_{V, j}) + \sum_a \mathcal{D}_V (\partial_{x_{V, a}} - A_{V, a}) + \sum_{b=1}^r \mathcal{D}_V (z_{V, b} \partial_{z_{V, b}} - \lambda_b) \right) ,$$

$$(20)$$

$$\mathbf{L}_{w, \theta}^!|_V \cong \mathcal{D}_V / \left(\sum_{j \in J_V} \mathcal{D}_V (q_{V, j} \partial_{q_{V, j}} - e_j - A_{V, j}) + \sum_a \mathcal{D}_V (\partial_{x_{V, a}} - A_{V, a}) + \sum_{b=1}^r \mathcal{D}_V (z_{V, b} \partial_{z_{V, b}} - \lambda_b) \right) .$$

Take $\xi_{V, j}, \eta_{V, a}$, and $\zeta_{V, b}$ as above. The principal symbols of the relations in both presentations (19), (20) are $(q_{V, j} \xi_{V, j})_{j \in J_V}, (\eta_{V, a})_a, (\zeta_{V, b})_{b=1}^r$. They form the regular sequence defining $\widetilde{\Sigma}_w \cap T^*V$.

By [Sai2, (3.10)], the trivial filtration $(F_k \mathbf{L}_{w, \theta}^{\geq -1} = \mathbf{L}_{w, \theta}^{\geq -1}$ when $k \geq 0$ and 0 when $k < 0$) induces the Hodge filtration:

$$F_k \mathbf{L}_{w, \theta} = \sum_{i+j=k} (F_i \mathcal{D}_{\widetilde{BS}_w}) (F_j \mathbf{L}_{w, \theta}^{\geq -1}) .$$

The analogous formula holds for $\mathbf{L}_{w,\theta}^!$ with $\mathbf{L}_w^{!>-1}$. Lemma 3.3 therefore gives after descent, nonequivariantly,

$$\begin{aligned} D_{\lambda\mathrm{gr}}(\mathbf{L}_{w,\theta}) &\cong (\iota_w)_* \mathcal{O}_{\Sigma_w}(E + P_\lambda), \\ D_{\lambda\mathrm{gr}}(\mathbf{L}_{w,\theta}^!) &\cong (\iota_w)_* \mathcal{O}_{\Sigma_w}(E + P_\lambda - E_\theta). \end{aligned}$$

Ad(B)-equivariance. Now we identify the $\mathrm{Ad}(B)$ equivariant structure of $D_{\lambda\mathrm{gr}}(\mathbf{L}_{w,\theta})$ and $D_{\lambda\mathrm{gr}}(\mathbf{L}_{w,\theta}^!)$ by computing the $\mathrm{Ad}(T)$ -weight of the rational section ψ from the proof of Lemma 3.3. Recall the isomorphism $U_w \times T \xrightarrow{\sim} \tilde{\mathcal{B}}_w$, $(u, t) \mapsto uwtU$. Moreover, $\psi|_{\tilde{\mathcal{B}}_w}$ is pullback of the section $1 \in \mathcal{K}_\lambda$ such that $\xi 1 = \langle \lambda, \xi \rangle 1$ for any $\xi \in \mathfrak{t}$. Therefore $\psi|_{\tilde{\mathcal{B}}_w}$ is invariant under the left, or equivalently adjoint U -action.

For $s \in T$ one has

$$s^{-1}(uwt)s = (s^{-1}us)w((w^{-1} \cdot s^{-1})ts),$$

Thus, with the action on sections defined by $(sf)(x) = f(s^{-1} \cdot x)$, we have

$$\xi \cdot \psi = \partial_{\xi - w^{-1}\xi} \psi = \langle \lambda, \xi - w^{-1}\xi \rangle \psi = \langle \lambda - w\lambda, \xi \rangle \psi.$$

For every SNC chart V , recall also the local sections $\phi_V = \prod_{j=1}^\ell q_{V,j}^{-n_j} \psi$ satisfying (15). Moreover, the coefficient of E_j in $E + P_\lambda$ is n_j . Therefore, B -equivariantly we have $\mathbf{L}_{w,\theta}^{\geq -1} \cong \mathcal{O}(D + F_\lambda) \otimes \mathbb{C}_{\lambda - w\lambda}$. Consequently,

$$D_{\lambda\mathrm{gr}}(\mathbf{L}_{w,\theta}) \cong (\iota_w)_* \mathcal{O}_{\Sigma_w}(E + P_\lambda) \otimes \mathbb{C}_{\lambda - w\lambda}.$$

Similarly,

$$D_{\lambda\mathrm{gr}}(\mathbf{L}_{w,\theta}^!) \cong (\iota_w)_* \mathcal{O}_{\Sigma_w}(E + P_\lambda - E_\theta) \otimes \mathbb{C}_{\lambda - w\lambda}.$$

Direct image. Finally we determine $D_{\lambda\mathrm{gr}}(\nabla_{w,\theta}^e)$ and $D_{\lambda\mathrm{gr}}(\Delta_{w,\theta}^e)$ by applying Theorem 2.1. Let $\tilde{j}_w : \tilde{\mathcal{B}}_w \hookrightarrow \widetilde{BS}_w$ and $j_w : \tilde{\mathcal{B}}_w \hookrightarrow \tilde{\mathcal{B}}$ be the open-cell and Bruhat-cell embeddings. The restriction $u := \tilde{p}|_{\tilde{\mathcal{B}}_w}$ is an isomorphism onto the Bruhat cell and $\tilde{p} \circ \tilde{j}_w = j_w \circ u$. Since $\mathbf{L}_{w,\theta} = (\tilde{j}_w)_\dagger \mathbb{L}_{w,\theta}$, functoriality gives

$$\tilde{p}_\dagger \mathbf{L}_{w,\theta} \cong (j_w)_\dagger u_\dagger \mathbb{L}_{w,\theta} \cong \nabla_{w,\theta}^e.$$

Similarly, using the properness of $\tilde{p}$,

$$\tilde{p}_\dagger \mathbf{L}_{w,\theta}^! \cong \tilde{p}_! (\tilde{j}_w)_! \mathbb{L}_{w,\theta} \cong (j_w)_! u_\dagger \mathbb{L}_{w,\theta} \cong \Delta_{w,\theta}^e.$$

After descent, Theorem 2.1 gives

$$D_{\lambda\mathrm{gr}}(\nabla_{w,\theta}^e) \cong \gamma_p D_{\lambda\mathrm{gr}}(\mathbf{L}_{w,\theta}), \quad D_{\lambda\mathrm{gr}}(\Delta_{w,\theta}^e) \cong \gamma_p D_{\lambda\mathrm{gr}}(\mathbf{L}_{w,\theta}^!),$$

where $\gamma_p(-) = (\varpi_p)_*(\rho_p^*(-) \otimes \omega_{BS_w/B} \langle b - \ell(w) \rangle)$. Set $\mathcal{T}_p := \omega_{BS_w/B} \otimes \mathcal{O}(E)$. Then equivariantly,

$$\mathcal{T}_p \cong \mathbb{C}_{-\rho} \otimes p^* \mathcal{O}_B(\rho).$$

Now the projection formula and the definitions of $\mathbf{K}_{w,\lambda}^\nabla$ and $\mathbf{K}_{w,\lambda}^\Delta$ give

$$\begin{aligned} D_{\lambda\mathrm{gr}}(\nabla_{w,\theta}^e) &\cong (\mathbf{K}_{w,\lambda}^\nabla)^e(\rho) \otimes \mathbb{C}_{\lambda - w\lambda - \rho} \langle b - \ell(w) \rangle, \\ D_{\lambda\mathrm{gr}}(\Delta_{w,\theta}^e) &\cong (\mathbf{K}_{w,\lambda}^\Delta)^e(\rho) \otimes \mathbb{C}_{\lambda - w\lambda - \rho} \langle b - \ell(w) \rangle. \end{aligned}$$

Finally we apply Proposition 2.3 to conclude the theorem. $\square$

3.5. Specializations. When λ satisfies certain conditions, the statements of Theorem 3.4 simplify, as we will see in Corollary 3.6.

For a subset $I \subset \{1, \dots, \ell\}$, put $E_I := \sum_{j \in I} E_j$ and define

$$(\mathbf{K}_{w,I})^e := (\varpi_p)_* (\rho_p^*((\iota_w)_* \mathcal{O}_{\Sigma_w}) \otimes \mathcal{O}(-E_I)),$$

where Σ_w is defined in the beginning of Section 3.4. Write $\mathbf{K}_{w,I} = \text{ind}_B^G(\mathbf{K}_{w,I})^e$. Recall (from e.g [BR, 1.2]) that the category $D^b \text{Coh}_{G \times G_m, Z}(\tilde{\mathcal{N}} \times \tilde{\mathcal{N}})$ comes equipped with the convolution:

$$(21) \quad \mathcal{F} \star \mathcal{G} := p_{13*}(p_{12}^* \mathcal{G} \otimes p_{23}^* \mathcal{F}).$$

Let ω denote the dualizing sheaf with its natural equivariant structure.

Lemma 3.5. *The following statements hold.*

- (1) $\mathbf{K}_{w,\emptyset} \cong \mathcal{O}_{\Lambda_w}$.
- (2) $\mathbf{K}_{w,\{1,\dots,\ell\}} \cong \omega_{\Lambda_w} \langle \ell(w) - b \rangle$.
- (3) $\mathbf{K}_{w,I} \cong M_{k_1}^I \star \dots \star M_{k_\ell}^I$, where

$$M_{k_j}^I := \begin{cases} \omega_{\Lambda_{s_{k_j}}} \langle 1 - b \rangle, & j \in I, \\ \mathcal{O}_{\Lambda_{s_{k_j}}}, & j \notin I. \end{cases}$$

Proof. Put $\Xi_w^e := A_w^e \times_{T^*BS_w}^R \Sigma_w$, where A_w^e is defined by (16). Let $q := \varpi_p|_{A_w^e}$. Then

$$(\varpi_p)_* \rho_p^*((\iota_w)_* \mathcal{O}_{\Sigma_w}) \cong q_* \mathcal{O}_{\Xi_w^e}.$$

Define

$$C_w^R := \Lambda_{s_{k_1}} \times_{\tilde{\mathcal{N}}}^R \dots \times_{\tilde{\mathcal{N}}}^R \Lambda_{s_{k_\ell}}.$$

The flags in C_w^R are ordered $(\mathbf{b}_0, \dots, \mathbf{b}_\ell)$. Let $C_w^{R,e}$ denote the fiber at $\mathbf{b}_0 = \mathbf{b}$. We first show there is a $B \times \mathbb{C}^\times$ -equivariant isomorphism of derived schemes

$$(22) \quad (\Xi_w^e =) A_w^e \times_{T^*BS_w}^R \Sigma_w \xrightarrow{\sim} C_w^{R,e}.$$

Let $\pi : T^*BS_w \rightarrow BS_w$ be the projection. The tautological one-form on T^*BS_w , followed by the natural inclusion $\Omega_{BS_w}^1 \hookrightarrow \Omega_{BS_w}^1(\log E)$, defines a section

$$\tau_{\log} \in \Gamma(T^*BS_w, \pi^* \Omega_{BS_w}^1(\log E)).$$

On an SNC chart with coordinates $t_1, \dots, t_r, y_1, \dots, y_m$ in which E is defined by $t_1 \cdots t_r = 0$ and dual cotangent coordinates $\eta_1, \dots, \eta_r, \zeta_1, \dots, \zeta_m$, the coefficients of $\tau_{\log}$ in the logarithmic basis are $t_1 \eta_1, \dots, t_r \eta_r, \zeta_1, \dots, \zeta_m$. Hence, $\Sigma_w = Z(\tau_{\log})$, the classical vanishing locus which in this case also coincides with the derived vanishing locus.

Pulling $\tau_{\log}$ back along $\rho_p : A_w^e \rightarrow T^*BS_w$ gives the logarithmic one-form

$$s_w := \rho_p^* \tau_{\log} \in \Gamma(A_w^e, \pi_A^* \Omega_{BS_w}^1(\log E)),$$

where $\pi_A : A_w^e \rightarrow BS_w$ is the projection. Derived zero loci (denoted by $Z^R(-)$) commute with base change, so

$$(23) \quad Z^R(s_w) \cong A_w^e \times_{T^*BS_w}^R \Sigma_w.$$

To show (22), we shall prove that $Z^R(s_w) \cong C_w^{R,e}$. We do so by induction along the Bott-Samelson tower. For $0 \leq j \leq \ell$, put

$$BS^{(j)} := P_{s_{k_1}} \times^B \dots \times^B P_{s_{k_j}} / B, \quad E^{(j)} := E_1 + \dots + E_j,$$

and let $\pi_j : BS^{(j)} \rightarrow BS^{(j-1)}$ forget the last flag. Since π_j is a smooth $\mathbb{P}^1$ -bundle and E_j is a section, there is an exact sequence

$$0 \rightarrow T_{BS^{(j)}/BS^{(j-1)}}(-\log E_j) \rightarrow T_{BS^{(j)}}(-\log E^{(j)}) \rightarrow \pi_j^* T_{BS^{(j-1)}}(-\log E^{(j-1)}) \rightarrow 0.$$

The first term has rank one. For $0 \leq j \leq \ell$, take

$$A_j = BS^{(j)} \times_{\mathcal{B}} T^* \mathcal{B} \cong \{(\mathfrak{b}_0, \dots, \mathfrak{b}_j, x), x \in \mathfrak{n}_{\mathfrak{b}_j}\}$$

such that $A_\ell = A_w^e$. Define a similar one-form $s_j \in \Gamma(A_j, a_j^* \Omega_{BS^{(j)}}^1(\log E^{(j)}))$. Trivialize the last minimal parabolic by choosing an $\mathfrak{sl}_2$ -triple (e, h, f) associated with the simple root α_{k_j} , with $e \in \mathfrak{n}_{\mathfrak{b}_{j-1}}$. Let u be the affine coordinate on the standard chart of $P_{s_{k_j}}/B \simeq \mathbb{P}^1$ parametrized by $\exp(uf)$ such that E_j is given by $u = 0$. Then affine-locally, $\mathfrak{b}_j(u) = \text{Ad}(\exp(uf))\mathfrak{b}_{j-1}$. Let ξ be the coordinate along the root space spanned by e . For $x \in \mathfrak{n}_{\mathfrak{b}_j}(u)$, we may write

$$x = x' + \xi \text{Ad}(\exp(uf))e = x' + \xi(e - uh - u^2 f)$$

where $x' \in \mathfrak{n}_{P_{s_{k_j}}}$. Then $x \in \mathfrak{n}_{\mathfrak{b}_{j-1}}$ is equivalent to $u\xi = 0$, which is the j -th $\mathbb{P}^1$ direction of s_j up to a unit. On the zero locus Υ_j of A_j , defined by $x \in \mathfrak{n}_{\mathfrak{b}_{j-1}}$, the form s_j descends to the quotient $(\pi_j^* T_{BS^{(j-1)}}(-\log E^{(j-1)}))|_{\Upsilon}$ to be the pullback of the section s_{j-1} for the truncated word. In all, we have

$$Z^R(s_j) \cong \Upsilon_j \times_{A_{j-1}}^R Z^R(s_{j-1}).$$

Since $\Upsilon_j \cong A_{j-1} \times_{\tilde{\mathcal{N}}} \Lambda_{s_{k_j}}$, induction on length therefore gives

$$Z^R(s_w) \cong \Lambda_{s_{k_1}} \times_{\tilde{\mathcal{N}}}^R \dots \times_{\tilde{\mathcal{N}}}^R \Lambda_{s_{k_\ell}}|_{\mathfrak{b}_0=\mathfrak{b}} (= C_w^{R,e}).$$

Together with (23), this proves (22).

By [BR, Theorem 2.2.1 and Remark 2.2.2(2)], $\mathcal{O}_{\Lambda_{s_{k_1}}} \star \dots \star \mathcal{O}_{\Lambda_{s_{k_\ell}}} \cong \mathcal{O}_{\Lambda_w}$. Thus part (1) follows.

For part (2), let $p_0, p_\ell : BS_w \rightarrow \mathcal{B}$ be the endpoint maps. We have the identity

$$(24) \quad \mathcal{K} := \omega_{BS_w/\mathcal{B}}(E) = p_0^* \mathcal{O}(\rho) \otimes p_\ell^* \mathcal{O}(-\rho).$$

Write $pr : A_w^e (= BS_w \times_{\mathcal{B}} T^* \mathcal{B}) \rightarrow BS_w$ for the projection. As seen in part (1), Ξ_w^e is the zero section of the rank $\ell(w)$ bundle $\mathcal{V} := pr^* \Omega_{BS_w/\mathcal{B}}^1(\log E)$. As a result, using (24) we have

$$\begin{aligned} \omega_{\Xi_w^e} &= (\omega_{A_w^e} \langle b \rangle \otimes \det \mathcal{V} \langle -\ell(w) \rangle)|_{\Xi_w^e} [\dim(A_w^e) - \text{rk}(\mathcal{V})] \\ &= \omega_{BS_w} \otimes p_\ell^* \omega_{\mathcal{B}} \otimes \omega_{BS_w/\mathcal{B}}(E) [\dim(A_w^e) - \text{rk}(\mathcal{V})] \\ &= p_0^* \omega_{\mathcal{B}} \otimes \mathcal{K}(-E) \otimes p_\ell^* \omega_{\mathcal{B}} \otimes \mathcal{K}[b] \\ &= \mathcal{O}_{\Xi_w^e}(-E) \langle b - \ell(w) \rangle [b]. \end{aligned}$$

Here the shift $\langle b \rangle$ on the first row comes from that A_w^e is of rank b over BS_w .

By part (1), $q_* \mathcal{O}_{\Xi_w^e} = \mathcal{O}_{\Lambda_w^e}$. Taking duality and using that Λ_w^e is Cohen-Macaulay by [BR, Theorem 2.2.1] give

$$q_* \mathcal{O}_{\Xi_w^e}(-E) \cong (i_w^e)_* \omega_{\Lambda_w^e} \langle \ell(w) - b \rangle$$

as desired.

For part (3), recall from (22) that

$$\Xi_w^e \cong \left(\Lambda_{s_{k_\ell}} \times_{\tilde{\mathcal{N}}}^R \dots \times_{\tilde{\mathcal{N}}}^R \Lambda_{s_{k_1}} \right) |_{\mathfrak{b}_0=\mathfrak{b}}.$$

Under this identification, the divisor E_j is the pullback of the boundary divisor in the one-step Bott–Samelson correspondence associated with s_{k_j} . Consequently,

$$\mathcal{O}_{\Xi_w^e}(-E_I) \cong \bigotimes_{j=1}^{\ell} \mathrm{pr}_j^* \mathcal{L}_j, \quad \mathcal{L}_j := \begin{cases} \mathcal{O}(-E_j), & j \in I, \\ \mathcal{O}, & j \notin I, \end{cases}$$

where pr_j denotes the projection to the j th one-step correspondence.

The endpoint pushforward of an iterated derived correspondence is the convolution of the endpoint pushforwards of its one-step factors. Applying derived base change and the projection formula successively along the Bott–Samelson tower therefore gives

$$Rq_* \mathcal{O}_{\Xi_w^e}(-E_I) \cong (M_{k_1}^I \star \cdots \star M_{k_\ell}^I)^e.$$

Indeed, if $j \notin I$, the one-step factor is $\mathcal{O}_{\Lambda_{s_{k_j}}}$ by part (1), while if $j \in I$, the one-step factor is $\omega_{\Lambda_{s_{k_j}}} \langle 1 - b \rangle$ by the length-one case of part (2). $\square$

The following result generalizes the case of $\theta = 1$ in [BIR].

Corollary 3.6. *Suppose $w \in W_\theta$ and $c_j \in [0, 1)$ for every j . We have the following identifications in $\mathrm{Coh}_{G \times \mathbb{G}_m}(\mathcal{Z}^R)$:*

- (1) $\mathrm{gr}_{\mathcal{H}}(\nabla_{w,\theta}) \cong \mathcal{O}_{\Lambda_w} \otimes \mathcal{O}(w\lambda - \lambda + \rho, \rho) \langle b - \ell(w) \rangle$.
- (2) $\mathrm{gr}_{\mathcal{H}}(\Delta_{w,\theta}) \cong (M_{k_1} \star \cdots \star M_{k_\ell}) \otimes \mathcal{O}(w\lambda - \lambda + \rho, \rho) \langle b - \ell(w) \rangle$ where

$$M_{k_j} = \begin{cases} \omega_{\Lambda_{s_{k_j}}} \langle 1 - \ell(w) \rangle, & \theta(\gamma_j) = 1, \\ \mathcal{O}_{\Lambda_{s_{k_j}}}, & \theta(\gamma_j) \neq 1. \end{cases}$$

Proof. The normalization gives $P_\lambda = 0$. Hence the corollary follows from Theorem 3.4 and Lemma 3.5. $\square$

4. TRACES OF LENGTH 0 ELEMENTS

In this section, take $G = \mathrm{SL}_n$ with Weyl group $W = S_n$. Let $W^{ex} = X^*(T) \rtimes W$ be the extended affine Weyl group.

4.1. Length 0 elements. For $1 \leq a \leq n$, let ω_a be the a -th fundamental weight. Put $\omega_0 = \omega_n = 0$ and $c \in S_n$ be the Coxeter element $s_1 \cdots s_{n-1}$. For any $\nu \in P$, denote by $t_\nu \in W^{ex}$ the translation by ν .

Lemma 4.1. *The length zero subgroup Ω of W^{ex} is generated by $t_{\omega_1} c$ with $(t_{\omega_1} c)^a = t_{\omega_a} c^a$ for all $1 \leq a \leq n$.*

Proof. The element $t_{\omega_1} c$ fixes the fundamental affine alcove and hence generates Ω . Moreover, $c^i t_{\omega_1} c^{-i} = t_{c^i(\omega_1)}$ and $\prod_{j=0}^{a-1} t_{c^j(\omega_1)} = t_{\omega_a}$, which gives the formula for its a -th power. $\square$

Since $c^a \rho/n - \rho/n = -\omega_a$, Corollary 3.6(1), applied with $\lambda = \rho/n$, gives the following special case. Its hypothesis holds for every $w \in W_\theta$: for any positive coroot $\beta^\vee$ one has $\langle \rho/n, \beta^\vee \rangle = \mathrm{ht}(\beta^\vee)/n \in [1/n, (n-1)/n] \subset [0, 1)$. The same calculation gives $e^{2\pi i \rho/n}(\beta^\vee) = e^{2\pi i \mathrm{ht}(\beta^\vee)/n} \neq 1$, so $D_\theta = 0$ for every $w \in W_\theta$.

Corollary 4.2. *For $\theta = e^{2\pi i \rho/n}$ and $1 \leq a \leq n$, $\Delta_{c^a, \theta} \cong IC_{c^a, \theta} \cong \nabla_{c^a, \theta}$. Moreover,*

$$D_{\lambda \mathrm{gr}}(\nabla_{c^a, e^{2\rho\pi\sqrt{-1}/n}}) \otimes \mathcal{O}(-\rho, -\rho) \cong (i_{c^a})_* \mathcal{O}_{\Lambda_{c^a}}(-\omega_a, 0) \langle b - a(n-a) \rangle.$$

It is easy to see that under Bezrukavnikov's two realization equivalence from [B] the right hand side indeed corresponds to the length 0 object in the affine Hecke category for PGL_n corresponding to a .

4.2. Cuspidal D -modules. Let $\mathcal{U}$ be the unipotent cone. We use the subscript r to denote regular locus. The regular unipotent orbit $\mathcal{U}_r \subset \mathcal{U}$ has fundamental group $\pi_1(\mathcal{U}_r) = \mathbb{Z}/n\mathbb{Z}$. For $0 \leq a \leq n-1$, let $\mathbb{K}_a$ be the integrable connection on $\mathcal{U}_r$ corresponding to $e^{2\pi a\sqrt{-1}/n} \in (\mathbb{Z}/n\mathbb{Z})^*$. Define the cuspidal D_G -module $\mathbb{N}_a$ to be the minimal extension of $\mathbb{K}_a$ to G . Equip $\mathbb{K}_a$ with the type-(0,0) Hodge structure. When a is coprime to n , $\mathbb{N}_a$ is clean, i.e, it is also the $\dagger$ - or $!$ -extension of $\mathbb{K}_a$ [Lu2, (17.8.4)]. For every $z \in Z_G$, let $z\mathbb{N}_a$ denote the pushforward of $\mathbb{N}_a$ under the multiplication map by z .

The following theorem is one of the main results of this paper.

Theorem 4.3. *When a is coprime to n , there is an isomorphism of Hodge modules*

$$\mathrm{CH}(\nabla_{c^a, e^{2\pi\rho\sqrt{-1}/n}}) \cong \bigoplus_{z \in Z_G} z\mathbb{N}_a(-m_a)$$

for $m_a = (n-1 + a(n-a))/2$, where the round brackets indicate the Tate twist.

Recall that $U_a = U \cap c^a \cdot U_-$. Let $S_a = U_{c^a} c^a$.

Lemma 4.4. *When a is coprime to n , the map $B/Z(G) \times S_a \rightarrow Bc^aB$, $(b, x) \mapsto b \cdot x$ is an isomorphism.*

Proof. To apply [NTY, Theorem 1.5], we first check the quasi-convexity condition, which we recall. Let $\Phi^\pm(w) := \{\alpha \in \Phi^\pm, w^m \alpha \in \Phi^\pm \text{ for all } m \in \mathbb{Z}\}$ and $\Phi(w) = \Phi^+(w) \sqcup \Phi^-(w)$. For a positive root α , define the sign change time by $n_w(\alpha) = \min\{m \geq 1, w^m \alpha \in \Phi^-\}$. Then the quasi-convexity condition is that $\Phi(w)$ is empty and $n_w(\alpha + \beta) \leq \max\{n_w(\alpha), n_w(\beta)\}$ for $\alpha, \beta \in \Phi^+$ such that $\alpha + \beta$ is a root. For $w = c^a$ and a positive root α_{ij} for $i < j$, as well as $1 \leq m < n$, let $q_m = n - (am \bmod n)$. Then $w^m \alpha_{ij}$ becomes negative exactly when $i \leq q - m \leq j$. Since a is coprime to n , we see that $\Phi(w)$ is empty. Moreover, for $i < j < k$, the first sign change time of $[i, k]$ is the minimum of those for $[i, j]$ and $[j, k]$ and hence the inequality holds. Therefore, [NTY, Theorem 1.5] implies that conjugation gives an isomorphism $U \times S_a \xrightarrow{\sim} U_{c^a} U$. Since c^a is an n -cycle, $\delta_{c^a} : T/Z(G) \rightarrow T$, $t \mapsto (c^a)^{-1} t c^a t^{-1}$, is also an isomorphism. The Bruhat factorization then identifies conjugation with an isomorphism $T/Z(G) \times U_{c^a} U \xrightarrow{\sim} Bc^aB$. Composing the two isomorphisms proves the claim. $\square$

Lemma 4.5. *Let $\theta = e^{2\pi\rho\sqrt{-1}/n}$ and let a be coprime to n . Then forgetting the Hodge module structure, $\mathrm{CH}(\nabla_{c^a, \theta})$ is a direct sum of shifts of copies of $\bigoplus_{z \in Z_G} z\mathbb{N}_a$.*

Proof. Recall the horocycle correspondence

$$(25) \quad G \xleftarrow{f} G \times^B G \xrightarrow{g} G \times^B \tilde{\mathcal{B}}.$$

Put $w = c^a$ and $K := \mathrm{CH}(\nabla_{w, \theta}) = f_! q^! \nabla_{w, \theta}$. Under the identification

$$G \times^B G \xrightarrow{\sim} G \times \mathcal{B}, \quad [g, h] \mapsto (ghg^{-1}, gB),$$

the map f becomes $f([g, h]) = ghg^{-1}$. By Lemma 4.4,

$$G \times^B (BwB) \cong G/Z_G \times S_a.$$

Using Corollary 4.2 and the smoothness of q , we see that $q^\dagger \nabla_{w,\theta}$ is the $!-, \dagger-$, and intermediate extension of an integrable connection $\mathbb{F}_a$ on $G/Z_G \times S_a$. Moreover, $\mathbb{F}_a$ is pulled back from the cover $G \rightarrow G/Z_G$ through the character χ_a of μ_n , $z \mapsto z^a$. Equip $\mathbb{F}_a$ with its weight-zero polarizable Hodge structure. Its intermediate extension is pure, and f is proper. Hence the decomposition theorem expresses K as a direct sum of shifts of simple equivariant pure Hodge modules.

The center acts on every stalk cohomology group of K through χ_a . For $x \in G$, let $d(x)$ be the gcd of its Jordan-block sizes. Then $\pi_0(C_G(x)) \cong \mu_{d(x)}$ and $\mu_n = Z_G \twoheadrightarrow \mu_{d(x)}$. The stabilizer action on a stalk factors through $\pi_0(C_G(x))$. Therefore a nonzero stalk forces χ_a to factor through $\mu_{d(x)}$. Since χ_a has order n , this is possible only when $d(x) = n$, i.e. when $x \in \cup_{Z_G} z\mathcal{U}_r$.

It follows that every simple summand of K is the intermediate extension of an irreducible equivariant local system on one of the orbits $z\mathcal{U}_r$. For $x \in z\mathcal{U}_r$, $\pi_0(C_G(x)) = Z_G = \mu_n$. Thus the central character determines the local system: after forgetting Hodge structures, it is $z\mathbb{K}_a$, and its intermediate extension is $z\mathbb{N}_a$.

Finally, the restriction of the T -action on Y to $Z_G \subset T$ corresponds to translation by Z_G on G . Hence CH intertwines these two actions. Since $\nabla_{c^a,\theta}$ is θ -monodromic, translation by any $z \in Z_G$ changes its equivariant structure only by the one-dimensional fiber of the character local system at z and therefore does not change its underlying object. It follows that $\text{CH}(\nabla_{c^a,\theta})$ is invariant under central translation and every summand $z\mathbb{N}_a$ occurs with the same shifts and multiplicities. $\square$

Proof of Theorem 4.3. We first show that $\text{CH}(\nabla_{c^a,\theta})$ is concentrated in cohomological degree 0. Use $\text{CH} = f_\dagger q^\dagger$ from the proof of Lemma 4.5. Consider the composition $\kappa : G \times^B (Bc^a B) \xrightarrow{j} G \times^B G \xrightarrow{f} G$. We have the clean integrable connection $\mathbb{F}_a$ on $G \times^B (Bc^a B)$ such that $\text{CH}(\nabla_{c^a,\theta}) = \kappa_\dagger \mathbb{F}_a = \kappa_! \mathbb{F}_a$, since f is proper. As κ is affine, we therefore have that $\text{CH}(\nabla_{c^a,\theta}) \in D^{\leq 0}(G) \cap D^{\geq 0}(G)$, where $D^{\leq 0}(G), D^{\geq 0}(G)$ stand for the nonpositive and nonnegative parts of the usual t-structure on the G -equivariant derived category of D-modules on G .

Thus Lemma 4.5 gives

$$\text{CH}(\nabla_{c^a,\theta}) \cong \left(\bigoplus_{z \in Z_G} z\mathbb{N}_a \right) \otimes Q_a$$

for a finite-dimensional Hodge structure Q_a . We determine Q_a from the stalk at a regular unipotent element x .

The fiber of κ at x is

$$X_x := \{(gZ_G, s) \in G/Z_G \times S_a \mid gsg^{-1} = x\}.$$

Projection to the second coordinate $\pi_x : X_x \rightarrow S_a \cap \mathcal{U}_r$ defines a torsor with fiber being $C_G(x)/Z_G \cong \mathbb{A}^{n-1}$.

Let $i_x : \{x\} \rightarrow G$ be the embedding. Proper base change gives

$$i_x^\dagger \text{CH}(\nabla_{c^a,\theta}) \cong R\Gamma_c(X_x, \text{DR}(\mathbb{F}_a|_{X_x})).$$

Under the de Rham functor, $\text{DR}(\mathbb{F}_a)|_{X_x} = L[\dim(X_x)]$ for some local system L . Let χ_c denote the compactly-supported Euler characteristic. Then $\chi_c(X_x, L) = \chi_c(X_x, \mathbb{C}_{X_x})$. By [Ki, Theorem 3.2], $\chi_c(S_a \cap \mathcal{U}_r, \mathbb{C}_{S_a \cap \mathcal{U}_r}) = 1$. Since $i_x^* \text{CH}(\nabla_{c^a,\theta})$ is concentrated in one cohomological degree, the component of that degree has to be of dimension 1. Thus $\dim(Q_a) = 1$.

Now the weight of $\mathbb{N}_a$ is $\dim(\mathcal{U}_r)$ and the weight of $j_+ \mathbb{F}_a$ is $\dim(G \times^B (Bc^a B))$. Therefore the weight of Q_a is

$$\dim(G \times^B (Bc^a B) - \dim(\mathcal{U}) = (\dim(G) + a(n - a)) - (\dim(G) - (n - 1)) = 2m_a.$$

We shall show that $\overline{Q_a} \cong Q_a$, which would imply that $Q_a \cong \mathbb{C}(-m_a)$. Since $\bar{\theta} = \theta^{-1}$, we have $\overline{\nabla_{c^a, \theta}} \cong \nabla_{c^a, \theta^{-1}}$, $\overline{\mathbb{N}_a} \cong \mathbb{N}_{n-a}$ and $\overline{Q_a} = Q_{n-a}$.

Let $\sigma : G \rightarrow G$ be the group automorphism induced by the Dynkin diagram involution. Then $\sigma(c) = c^{-1}$ and $\sigma(z) = z^{-1}$ for $z \in Z(G)$. Therefore,

$$\sigma^* \nabla_{c^a, \theta^{-1}} \cong \nabla_{c^{n-a}, \theta}, \quad \sigma^* \mathbb{N}_{n-a} \cong \mathbb{N}_a.$$

Consequently $\overline{Q_a} \cong Q_a$ as Hodge structures and the theorem follows. $\square$

Corollary 4.6. *Let a be coprime to n , put $\theta = e^{2\pi\rho\sqrt{-1}/n}$. Then in $\mathrm{Coh}_{\mathcal{N}, G \times \mathbb{C}^\times}(\mathfrak{C})$,*

$$\widetilde{\mathrm{CH}}(\mathrm{gr}_{\mathcal{H}}(\nabla_{c^a, \theta})) \cong \mathrm{gr}_{\mathcal{CS}} \left(\bigoplus_{z \in Z_G} z \mathbb{N}_a(-m_a) \right).$$

Corollary 3.6 identifies the left hand side as $\widetilde{\mathrm{CH}}((i_{c^a})_* \mathcal{O}_{c^a}(-\omega_a + \rho, \rho)) \langle b - a(n - a) \rangle$.

4.3. The GL_n lift. Firstly, for GL_n , the length-zero subgroup is $\mathbb{Z}$ generated by $\varpi := t_{\omega_1} c$ with

$$(26) \quad \varpi^n = t_\zeta, \quad \zeta := (1, 1, \dots, 1) \in X^*(T).$$

We first make the descent from SL_n explicit. Put

$$\mathcal{K}_{a,n}^{\mathrm{SL}} := \mathrm{gr}_{\mathcal{H}}(\nabla_{c^a, \theta}), \quad \overline{\mathbb{N}}_{a,n}^{\mathrm{SL}} := \bigoplus_{z \in \mu_n} z \mathbb{N}_a.$$

Consider the central isogeny $\pi : \mathrm{SL}_n \times \mathbb{C}^\times \rightarrow \mathrm{GL}_n$, $(g, t) \mapsto tg$ whose kernel is the antidiagonal μ_n . Take $\overline{\mathcal{Z}}^R := \widetilde{\mathcal{N}} \times_{\mathfrak{gl}_n}^R \widetilde{\mathcal{N}}$. Under the decomposition $\mathfrak{gl}_n = \mathfrak{sl}_n \oplus \mathfrak{z}$, we have

$$\overline{\mathcal{Z}}^R = \mathcal{Z}^R \times \mathrm{Spec}(\mathrm{Sym}(\mathfrak{z}^*[1])).$$

Let $q : \mathcal{Z}^R \rightarrow \overline{\mathcal{Z}}^R$ denote the quotient map.

Write $d = a + kn$ with $1 \leq a \leq n - 1$ and a coprime to n when $n > 1$ and set $a = 0$ when $n = 1$. Define

$$\mathcal{K}_{d,n} := q_* \mathcal{K}_a^{\mathrm{SL}} \otimes \mathbb{C}_{\det^k}, \quad \overline{\mathbb{N}}_{d,n} := \pi_* \left(\overline{\mathbb{N}}_{a,n}^{\mathrm{SL}} \boxtimes \mathcal{O}_{\mathbb{C}^\times} \right)^{\chi_a} \otimes \mathbb{C}_{\det^k}.$$

On $\mathcal{K}_{a,n}^{\mathrm{SL}}$, we define a scaling $\mathbb{C}^\times$ -equivariance by letting $t \in \mathbb{C}^\times$ act on local sections as multiplication by t^a . Then the $\mathrm{SL}_n \times \mathbb{C}^\times$ -equivariance on $\mathcal{K}_{d,n}$ descends along π to a GL_n -equivariant structure. Similar for $\overline{\mathbb{N}}_{d,n}$. Corollary 4.6 plus μ_n -equivariance implies the following:

Corollary 4.7. *In $\mathrm{Coh}_{\mathcal{N}, \mathrm{GL}_n \times \mathbb{C}^\times}(\mathfrak{C}_n)$ there is an isomorphism*

$$\widetilde{\mathrm{CH}}_n(\mathcal{K}_{d,n}) \cong \mathrm{gr}_{\mathcal{CS}}(\overline{\mathbb{N}}_{d,n}(-m_a)).$$

5. COMPATIBILITY WITH PARABOLIC INDUCTION

5.1. Parabolic induction functors. Here we consider the following situation. Let G be a connected reductive group, $P \subset G$ be a parabolic subgroup, and $L \subset P$ be a Levi subgroup. Choose a maximal torus $T \subset L$ and let θ have the same meaning as in Section 2.3. We can form the following spaces and categories:

- (i) We can consider the derived commuting schemes $\mathfrak{C}_L, \mathfrak{C}_G$ for L and G . Similarly define $\mathfrak{C}_P$ to be the derived zero fiber of the map $P \times \mathfrak{p} \rightarrow \mathfrak{p}, (p, x) \mapsto \text{Ad}(p)x - x$. There is a correspondence

$$\mathfrak{C}_L/L \xleftarrow{\iota} \mathfrak{C}_P/P \xrightarrow{\pi} \mathfrak{C}_G/G,$$

where ι is induced by the projection $P \twoheadrightarrow L$ and π is induced by the inclusion $P \hookrightarrow G$. This correspondence gives rise to the pull-push functor as follows. Let ω denote the canonical bundle of P with its natural P -equivariant structure that we lift to $\mathfrak{C}_P/P$ and denote by the same letter. Then we consider the functor

$$\pi_* \iota^* : \text{Coh}_{L \times \mathbb{C}^\times}(\mathfrak{C}_L) \rightarrow \text{Coh}_{G \times \mathbb{C}^\times}(\mathfrak{C}_G)$$

that restricts to

$$\text{Coh}_{\mathcal{N}, L \times \mathbb{C}^\times}(\mathfrak{C}_L) \rightarrow \text{Coh}_{\mathcal{N}, G \times \mathbb{C}^\times}(\mathfrak{C}_G).$$

We denote this functor by ${}^c\widetilde{\text{Ind}}_P^G$.

Similarly to Section 2.6, the functor ${}^c\widetilde{\text{Ind}}_P^G$ decomposes as the composition of two functors between categories of coherent sheaves on cotangent bundles introduced in Section 2.2. First we apply the pull-push along

$$T^*(L/\text{Ad}(L)) \leftarrow P/\text{Ad}(P) \times_{L/\text{Ad}(L)} T^*(L/\text{Ad}(L)) \rightarrow T^*(P/\text{Ad}(P))$$

that is associated to the smooth morphism $P/\text{Ad}(P) \rightarrow L/\text{Ad}(L)$ and then the twisted (by the canonical bundle of $P/\text{Ad}(P)$) pull-push via

$$T^*(P/\text{Ad}(P)) \leftarrow P/\text{Ad}(P) \times_{G/\text{Ad}(G)} T^*(G/\text{Ad}(G)) \rightarrow T^*(G/\text{Ad}(G)).$$

It is associated to the projective morphism $P/\text{Ad}(P) \rightarrow G/\text{Ad}(G)$.

- (i') Similarly, we have the parabolic induction functor

$${}^c\text{Ind}_P^G : D_L^b(L) \rightarrow D_G^b(G).$$

- (ii) We can form the Steinberg derived schemes $\mathcal{Z}_L, \mathcal{Z}_G$, as well as the (derived) affine Hecke categories $\mathcal{AH}_L, \mathcal{AH}_G$. We can view these categories as the dg categories $\text{Coh}_{\mathbb{C}^\times}(T^*(B_L \setminus L/B_L)), \text{Coh}_{\mathbb{C}^\times}(T^*(B \setminus G/B))$, where $B_L := L \cap B$. We can then, as in

- (i), define the composition ${}_{\mathcal{H}}\widetilde{\text{Ind}}_P^G$ of pull-push functors associated to stack morphisms

$$B_L \setminus L/B_L \leftarrow B \setminus P/B \rightarrow B \setminus G/B.$$

- (ii') Similarly to (i'), we have the categories $\mathcal{H}_{L,\theta}, \mathcal{H}_{G,\theta}$ and the induction functor

$${}_{\mathcal{H}}\text{Ind}_P^G : \mathcal{H}_{L,\theta} \rightarrow \mathcal{H}_{G,\theta}.$$

Remark 5.1. If Z_G has positive dimension (for instance, when $G = \text{GL}_n$), the categories above should be understood within the χ -twisted equivariant framework introduced in Remark 2.5. The character $\chi : \mathfrak{g} \rightarrow \mathbb{C}$ restricts naturally along $\mathfrak{l} \subset \mathfrak{p} \subset \mathfrak{g}$, and the four induction operations preserve this twist; hence, we omit explicit mention of χ .

5.2. Compatibility of functors.

Lemma 5.2. *For all θ , the functor ${}_c\mathrm{Ind}_P^G$ sends $\mathcal{CS}_{L,\theta}$ to $\mathcal{CS}_{G,\theta}$. Moreover, the following diagram is commutative, where*

- (1) *the vertical arrows are the functors of taking the Hodge associated graded sheaf;*
- (2) *the horizontal arrows are the character functors CH or $\widetilde{\mathrm{CH}}$;*
- (3) *and the slant arrows are parabolic induction functors ${}_?\mathrm{Ind}_P^G$ or ${}_?\widetilde{\mathrm{Ind}}_P^G$.*

$$\begin{array}{ccccc}
 \mathcal{H}_{L,\theta} & \xrightarrow{\quad} & \mathcal{CS}_{L,\theta} & & \\
 \downarrow & \searrow & \downarrow & \searrow & \\
 & \mathcal{H}_{G,\theta} & \xrightarrow{\quad} & \mathcal{CS}_{G,\theta} & \\
 \downarrow & \downarrow & \downarrow & \downarrow & \\
 \mathcal{AH}_L & \xrightarrow{\quad} & \mathrm{Coh}_{L \times \mathbb{C}^\times, \mathcal{N}}(\mathfrak{C}_L) & \searrow & \\
 & \downarrow & \downarrow & \downarrow & \\
 & \mathcal{AH}_G & \xrightarrow{\quad} & \mathrm{Coh}_{G \times \mathbb{C}^\times, \mathcal{N}}(\mathfrak{C}_G) &
 \end{array}$$

(27)

Proof. 1) Induction vs character functors.

We start with the commutativity of the version of the top square of (27), where instead of the categories of character sheaves we have $D_L^b(L)$ and $D_G^b(G)$. Consider the following commutative diagram of quotient stacks:

$$\begin{array}{ccccc}
 L/\mathrm{Ad} L & \longleftarrow & P/\mathrm{Ad} P & \longrightarrow & G/\mathrm{Ad} G \\
 \uparrow & & \uparrow & & \uparrow \\
 L/\mathrm{Ad} B_L & \longleftarrow & P/\mathrm{Ad} B & \longrightarrow & G/\mathrm{Ad} B \\
 \downarrow & & \downarrow & & \downarrow \\
 Y_L & \longleftarrow & Y_P & \longrightarrow & Y_G
 \end{array}$$

(28)

where

$$Y_L = (U_L \backslash L / U_L) / T, Y_P = (U \backslash P / U) / T, Y_G = (U \backslash G / U) / T.$$

We note that

- (i) The composition ${}_c\mathrm{Ind}_P^G \circ \mathrm{CH}_L$ is the composition of the pull-push along the leftmost column followed by the pull-push along the top row.
- (ii) The composition $\mathrm{CH}_G \circ {}_{\mathcal{H}}\mathrm{Ind}_P^G$ is the composition of the pull-push along the bottom row followed by the pull-push along the right column.
- (iii) The top-left and bottom-right squares in the diagram are Cartesian.
- (iv) The morphisms $Y_P \rightarrow Y_G$ and $L/\mathrm{Ad} B_L \rightarrow L/\mathrm{Ad} L$ are proper.

So we can apply proper base change and conclude that the functors in (i) and (ii) are both isomorphic to the pull-push via

$$Y_L \leftarrow P/\mathrm{Ad} B \rightarrow G/\mathrm{Ad} G$$

This shows the commutativity of the top square of (27) modified as in the beginning of the proof and also that ${}_c\mathrm{Ind}_P^G$ sends objects in the image of CH_L to $\mathcal{CS}_{G,\theta}$. For the assertion about the image of $\mathcal{CS}_{L,\theta}$ under ${}_c\mathrm{Ind}_P^G$, by [BYD], ${}_c\mathrm{Ind}_P^G$ is t -exact. Exactness then preserves subquotients and extensions and hence the Serre subcategory $\mathcal{CS}_{L,\theta}^\heartsuit$ and implies that ${}_c\mathrm{Ind}_P^G \mathcal{CS}_{L,\theta}^\heartsuit$ lands in $\mathcal{CS}_{G,\theta}$.

Now we proceed to the commutativity of the bottom square of (27). In order to establish this we need the following commutative diagram completely analogous to (28):

$$\begin{array}{ccccc} L/\mathrm{Ad} L & \longleftarrow & P/\mathrm{Ad} P & \longrightarrow & G/\mathrm{Ad} G \\ \uparrow & & \uparrow & & \uparrow \\ L/\mathrm{Ad} B_L & \longleftarrow & P/\mathrm{Ad} B & \longrightarrow & G/\mathrm{Ad} B \\ \downarrow & & \downarrow & & \downarrow \\ B_L \backslash L/B_L & \longleftarrow & B \backslash P/B & \longrightarrow & B \backslash G/B \end{array}$$

Note that this diagram satisfies analogs of (iii) and (iv) above. So we can apply the proper base change for functors between categories of coherent sheaves on cotangent bundles, cf. Section 2.6. This yields the commutativity of the bottom square of (27) following the scheme explained above for the D-module functors, cf. [BeLo, Lemma 4.6].

2) Character vs Hodge associated graded functors: this is the content of Theorem 2.8.

3) Induction vs Hodge associated graded functors: this is completely analogous to 2) and is left as an exercise. $\square$

5.3. Monoidal structure. We consider the following situation. Let $Q \subset G$ be a parabolic containing P and $M \subset Q$ a Levi subgroup containing L . Write P_M for $P \cap M$, this is a parabolic subgroup. Proper base change yields the following isomorphisms of functors

$$(29) \quad \psi_{Q,P}^G : {}_c\widetilde{\mathrm{Ind}}_Q^G \circ {}_c\widetilde{\mathrm{Ind}}_{P_M}^M \xrightarrow{\sim} {}_c\widetilde{\mathrm{Ind}}_P^G.$$

Now suppose we are given three embedded parabolic subgroups $P \subset Q \subset R \subset G$ with Levi subgroups $L \subset M \subset K$. The notation like P_K is used for the intersection like $K \cap P$. We have the following transitivity result.

Lemma 5.3. *We have an equality*

$$\psi_{R,P}^G \circ (1\psi_{Q_K,P_K}^K) = \psi_{Q,P}^G \circ (\psi_{R,Q}^G 1).$$

of functor morphisms

$${}_c\widetilde{\mathrm{Ind}}_R^G \circ {}_c\widetilde{\mathrm{Ind}}_{Q_K}^K \circ {}_c\widetilde{\mathrm{Ind}}_{P_M}^M \xrightarrow{\sim} {}_c\widetilde{\mathrm{Ind}}_P^G$$

Here 1 denotes the identity endomorphisms of the appropriate functors and the notation like $1\psi_{Q_K,P_K}^K$ denotes the natural transformation between the corresponding composite functors.

Proof. Consider the following commutative diagram.

$$\begin{array}{ccccc}
 & & \mathfrak{C}_P/P & & \\
 & \swarrow & & \searrow & \\
 & \mathfrak{C}_{P_K}/P_K & & \mathfrak{C}_Q/Q & \\
 & \swarrow & & \searrow & \\
 \mathfrak{C}_{P_M}/P_M & & \mathfrak{C}_{Q_K}/Q_K & & \mathfrak{C}_R/R \\
 \swarrow & & \searrow & & \swarrow \\
 \mathfrak{C}_L/L & & \mathfrak{C}_M/M & & \mathfrak{C}_K/K \\
 & & & & \searrow \\
 & & & & \mathfrak{C}_G/G
 \end{array}$$

The three squares are derived Cartesian: take the upper square as an example. We have that $\mathfrak{C}_P \cong (P \times \mathfrak{p}) \times_{\mathfrak{p}}^R \{0\}$ and

$$(30) \quad \mathfrak{C}_P/\mathrm{Ad}(P) \cong [(P \times \mathfrak{p})/\mathrm{Ad}(P)] \times_{[\mathfrak{p}/\mathrm{Ad}(P)]}^R \mathbf{B}P,$$

where we write $\mathbf{B}P$ for pt/P , while $(P \times \mathfrak{p})/\mathrm{Ad}(P) \cong P/\mathrm{Ad}(P) \times_{\mathbf{B}P} \mathfrak{p}/\mathrm{Ad}(P)$.

Let $\mathfrak{q} := \mathrm{Lie}(Q)$. We have isomorphisms $P \cong P_K \times_{Q_K} Q$ and $\mathfrak{p} \cong \mathfrak{p}_K \times_{\mathfrak{q}_K} \mathfrak{q}$. Since the maps $Q \rightarrow Q_K$ and $\mathfrak{q} \rightarrow \mathfrak{q}_K$ are smooth and surjective, we have $P_K \times_{Q_K} Q \cong P_K \times_{Q_K}^R Q$, $\mathfrak{p}_K \times_{\mathfrak{q}_K} \mathfrak{q} = \mathfrak{p}_K \times_{\mathfrak{q}_K}^R \mathfrak{q}$. We also have $\mathbf{B}P \cong \mathbf{B}P_K \times_{\mathbf{B}Q_K}^R \mathbf{B}Q$. Moreover, taking loop stacks, resp. shifted tangent stack, preserves derived fiber products and gives $L(\mathbf{B}P) \cong [P/\mathrm{Ad}(P)]$, resp. $T_{\mathbf{B}P}[-1] \cong [\mathfrak{p}/\mathrm{Ad}(P)]$. Hence we also have

$$[P/\mathrm{Ad}(P)] \cong [P_K/\mathrm{Ad}(P_K)] \times_{[Q_K/\mathrm{Ad}(Q_K)]}^R [Q/\mathrm{Ad}(Q)],$$

$$[\mathfrak{p}/\mathrm{Ad}(P)] \cong [\mathfrak{p}_K/\mathrm{Ad}(P_K)] \times_{[\mathfrak{q}_K/\mathrm{Ad}(Q_K)]}^R [\mathfrak{q}/\mathrm{Ad}(Q)].$$

Finally, given the description (30), commutation of finite limits proves the upper square is derived Cartesian.

The isomorphisms $\psi_{R,P}^G \circ (1\psi_{Q_K,P_K}^K)$ and $\psi_{Q,P}^G \circ (\psi_{R,Q}^G 1)$ are composition of base changes through the three derived Cartesian squares in the diagram above, and therefore are equal. $\square$

Now suppose $G = \mathrm{GL}_n$. We write G_n for GL_n , $\mathfrak{C}_n$ for the corresponding commuting derived scheme, and $\tilde{\mathfrak{C}}_n$ for $\mathrm{Coh}_{\mathcal{N}, G_n \times \mathbb{C}^\times}(\mathfrak{C}_n)$. Our task is to equip $\bigoplus_{n \geq 0} \tilde{\mathfrak{C}}_n$ with a monoidal structure. For a composition $\vec{n} = (n_1, \dots, n_k)$ we write $P_{\vec{n}}$ for the stabilizer of the standard partial flag $\mathbb{C}^{n_1} \subset \mathbb{C}^{n_1+n_2} \dots \mathbb{C}^{n_1+\dots+n_i} \subset \dots \subset \mathbb{C}^n$ where $n = \sum_i n_i$.

- 1) The tensor product functor $\star : \tilde{\mathfrak{C}}_n \boxtimes \tilde{\mathfrak{C}}_m \rightarrow \tilde{\mathfrak{C}}_{n+m}$ is given by ${}_c\widetilde{\mathrm{Ind}}_{P_{(n,m)}}^{G_{n+m}}$.
- 2) The associativity isomorphism

$$(\mathcal{F}_1 \star \mathcal{F}_2) \star \mathcal{F}_3 \xrightarrow{\sim} \mathcal{F}_1 \star (\mathcal{F}_2 \star \mathcal{F}_3), \mathcal{F}_i \in \tilde{\mathfrak{C}}_{n_i},$$

is given by

$$(\psi_{(n_1, n_2, n_3), \vec{n}}^n)^{-1} \circ \psi_{(n_1+n_2, n_3), \vec{n}}^n,$$

where $\vec{n} = (n_1, n_2, n_3)$ and the compositions in the subscript indicate the corresponding parabolics.

- 3) The monoidal unit is the object $\mathbb{C}$ in $\tilde{\mathfrak{C}}_0 = D^b\mathrm{Rep}(\mathbb{C}^\times)$.

4) The pentagon axiom is a consequence of Lemma 5.3: under the identification of all terms involved with $\mathrm{Ind}_{\vec{n}}^n(\mathcal{F}_1 \boxtimes \mathcal{F}_2 \boxtimes \mathcal{F}_3 \boxtimes \mathcal{F}_4)$ all isomorphisms in the pentagon diagram become the identity, as the lemma shows that these isomorphisms do not depend on the way to bracket.

Remark 5.4. Note that we have analogs of (29) for the affine Hecke categories for GL_n . We write $\mathcal{AH}_n$ for the affine Hecke category for GL_n . Direct analogs of Lemma 5.3 can be used to show that

- $\bigoplus_{n \geq 0} \mathcal{AH}_n$ is a monoidal category with respect to the parabolic induction.
- And $\bigoplus_{n \geq 0} \widetilde{\mathrm{CH}}$ is a monoidal functor.

6. SYMMETRIC GROUP ACTIONS

Suppose now $n = dr$ for $d, r \in \mathbb{Z}_{>0}$. Set $L = \mathrm{GL}_r^d \subset G = \mathrm{GL}_n$ with an associated parabolic P . Recall the D -module $\mathbb{N}_{a,r}$ on GL_r from Section 4.3. We may consider the convolution power

$$(31) \quad \overline{\mathbb{N}}_{a,r}^{\star d} := {}_c\mathrm{Ind}_P^{G \times \mathbb{A}^d} \overline{\mathbb{N}}_{a,r}^{\boxtimes d}$$

for $1 \leq a \leq r-1$ coprime to r when $r > 1$ and $a = 0$ when $r = 1$. The main result in this section is the following.

Theorem 6.1. *There is an injection $\mathbb{C}[S_d] \hookrightarrow \mathrm{End}_{\mathrm{MHM}(G)}(\overline{\mathbb{N}}_{a,r}^{\star d})$.*

Note that when $r = 1$, $\overline{\mathbb{N}}_{0,1}^{\star n}$ is the group version Grothendieck-Springer D -module, and the result is standard. Our construction generalizes this case and follows a similar proof strategy as in [HK].

To set up the proof, let $W_L = S_r^d$ be the Weyl group of the Levi $L = \mathrm{GL}_r^d$; the relative Weyl group $N_W(W_L)/W_L$ is S_d , permuting the d blocks, and it is this S_d that will act below. Take $T_d = Z(L)$ and put $A_d := P/[P, P] \cong L/[L, L]$. Let $\nu : T_d \rightarrow A_d$ be the natural isogeny. Let

$$T_d^\circ := \{t \in T_d \mid \mathrm{Stab}_W(t) = W_L, \mathrm{Stab}_{S_d}(\nu(t)) = \{1\}\}$$

and set $A_d^\circ := \nu(T_d^\circ)$.

Let U_P be the unipotent radical of P , and let $O_{(r)} \subset \mathrm{GL}_r$ be the principal unipotent orbit. Take

$$\tilde{G}_d^\circ := G \times^P (T_d^\circ \cdot O_{(r)}^d \cdot U_P).$$

There is an isomorphism

$$(32) \quad U_P \times T_d \times O_{(r)}^d \rightarrow T_d \cdot O_{(r)}^d \cdot U_P, \quad (u, t, x) \mapsto u \cdot (tx).$$

Define $G_d^\circ \subset G$ to be the image of $\tilde{G}_d^\circ$ under $G \times^P P \rightarrow G$, $[g, p] \mapsto g \cdot p$. There is an embedding

$$\tilde{G}_d^\circ \rightarrow G \times A_d, \quad [g, tuv] \mapsto (g \cdot (tuv), \nu(t))$$

Moreover, $\tilde{G}_d^\circ \cong G_d^\circ \times_{A_d^\circ/S_d} A_d^\circ$, so the induced map $\tilde{G}_d^\circ \rightarrow G_d^\circ$ is an S_d -Galois cover. Set G_d to be the closure of G_d° in G .

Recall that $\pi_1(O_{(r)}) = \mathbb{Z}/r\mathbb{Z}$. Let $\mathbb{K}_a$ be the integrable connection on $O_{(r)}$ corresponding to $e^{2\pi a \sqrt{-1}/r} \in (\mathbb{Z}/r\mathbb{Z})^*$. Let $\tilde{\mathbb{E}}_a^e$ be the connection on $T_d \cdot O_{(r)}^d \cdot U_P$ pulled back from the connection $\mathbb{K}_a^{\boxtimes d}$ on $O_{(r)}^d$ with respect to (32). Put $S(P) = P \cap \mathrm{SL}_n$ and $S(L) = L \cap \mathrm{SL}_n$. Each $\mathbb{K}_a$ is GL_r -twisted equivariant. Moreover, $Z(\mathrm{GL}_r)$ acts on $\mathbb{K}_a$ by $z \mapsto z^a$. Therefore, $\mathbb{K}_a^{\boxtimes d}$ is $S(L)$ -equivariant.

Since a and r are coprime, each component of this character has order r . Let $j : T_d \cdot O_{(r)}^d \cdot U_P \hookrightarrow P$ be the locally closed inclusion. In each GL_r block, the restriction of $\tilde{\mathbb{E}}_a^e$ is clean as in Section 4. Therefore, $j_! \tilde{\mathbb{E}}_a^e \cong j_{!*} \tilde{\mathbb{E}}_a^e \cong j_+ \tilde{\mathbb{E}}_a^e$, and we denote this object by $\tilde{\mathbb{M}}_a^e$. Let

$\widetilde{\mathbb{M}}_a = \text{Ind}_{S(P)}^{\text{SL}_n} \widetilde{\mathbb{M}}_a^e$, this is a D-module on $G \times^P P$. It is also the $\dagger$ -extension of $\widetilde{\mathbb{E}}_a := \text{Ind}_{S(P)}^{\text{SL}_n} \widetilde{\mathbb{E}}_a^e$.

By construction, $\overline{\mathbb{N}}_{a,r}^{*d}$ is the pushforward of $\widetilde{\mathbb{M}}_a$ to G .

Let $\mathcal{U}_L$ denote the unipotent variety of L , $\mathcal{N}_L$ denote the nilpotent cone in $\mathfrak{l}$. Under the trace form, we identify $\mathfrak{g} \cong \mathfrak{g}^*$ as well as $\mathfrak{p}^* \cong \mathfrak{p}_-$ etc.

Lemma 6.2. *We have $SS(\widetilde{\mathbb{M}}_a) = G \times^P \Sigma$, where*

$$\Sigma = \{(tuv, x) \mid t \in T_d, u \in \mathcal{U}_L, v \in U_P, x \in \mathcal{N}_L, u \cdot x = x\}.$$

Proof. First of all, by [Lu, Theorem 5(b)], $SS(\mathbb{N}_{a,r}) = \{(u, x) \mid u \in \mathcal{U}_{\text{GL}_r}, x \in \mathcal{N}_{\text{GL}_r}, u \cdot x = x\}$. As a result,

$$SS(\mathbb{N}_{a,r}) = \{(zu, x) \mid z \in \mathbb{C}^\times, u \in \mathcal{U}_{\text{GL}_r}, x \in \mathcal{N}_{\text{GL}_r}, u \cdot x = x\}.$$

Hence by the block-wise construction, $SS(\widetilde{\mathbb{M}}_a^e) = \Sigma$. The lemma now follows from Proposition 2.3. $\square$

6.1. Generalized Springer D-module. Let $q_P : P \longrightarrow A_d = P/[P, P]$ be the quotient map and define

$$f : G \times^P P \longrightarrow G \times A_d, \quad [g, h] \longmapsto (g \cdot h, q_P(h)).$$

In particular, if $h = tuv$ with $t \in T_d$, $u \in \mathcal{U}_L$, and $v \in U_P$, then $q_P(h) = \nu(t)$.

Define $\mathbb{M}_a := f_+ \widetilde{\mathbb{M}}_a$, which resembles the Harish-Chandra D -module of Hotta–Kashiwara [HK]. Let $\mathfrak{t}_d := \text{Lie}(T_d)$ such that $\mathfrak{p}$ has a splitting $\mathfrak{t}_d \oplus [\mathfrak{p}, \mathfrak{p}]$ and hence $\text{Lie}(A_d) \cong \mathfrak{t}_d$.

We identify

$$T^*(G \times A_d) \cong G \times A_d \times \mathfrak{g} \times \mathfrak{t}_d.$$

Lemma 6.3. *The singular support of $\mathbb{M}_a$ is $\Lambda := \overline{G \cdot \Lambda_L}$, where*

$$\Lambda_L := \{(tu, \nu(t), x + y, -y) \mid t \in T_d, u \in \mathcal{U}_L, x \in \mathcal{N}_L, y \in \mathfrak{t}_d, u \cdot x = x\}.$$

Moreover, Λ is irreducible and Lagrangian in $T^(G \times A_d)$.*

Proof. Take

$$i : G \times^P P \longrightarrow G/P \times G \times A_d, \quad [g, h] \longmapsto (gP, g \cdot h, q_P(h)).$$

Recall the maps ρ_i and ϖ_i from (1). Since i is a closed embedding,

$$SS(i_+ \widetilde{\mathbb{M}}_a) = \varpi_i \rho_i^{-1}(SS(\widetilde{\mathbb{M}}_a)).$$

By G -equivariance, it suffices to calculate $SS(i_+ \widetilde{\mathbb{M}}_a)$ at $[1, h]$. For $\alpha \in \mathfrak{g}$ and $\beta \in \mathfrak{p}$, we have

$$di_{[1, h]}(\alpha, \beta) = (\alpha \bmod \mathfrak{p}, h^{-1} \cdot \alpha - \alpha + \beta, dq_P(\beta)).$$

Suppose that

$$(x, y, z) \in T_{i([1, h])}^*(G/P \times G \times A_d)$$

annihilates the image of $di_{[1, h]}$. Then

$$\langle x, \alpha \rangle + \langle y, h^{-1} \cdot \alpha - \alpha + \beta \rangle + \langle z, dq_P(\beta) \rangle = 0$$

for every $\alpha \in \mathfrak{g}$ and $\beta \in \mathfrak{p}$. It follows that $x = y - h \cdot y$,

$$y \in [\mathfrak{p}, \mathfrak{p}]^\perp = \mathfrak{t}_d + \mathfrak{n}_P, \quad z = -y.$$

Hence

$$\text{Ker}(d^* i_{[1, h]}) = \{(y - h \cdot y, y, -y) \mid y \in \mathfrak{t}_d + \mathfrak{n}_P\}.$$

Given Lemma 6.2, we obtain

$$SS(i_+^* \widetilde{\mathbb{M}}_a)|_{i([1, tuv])} = \{(x + y - tuv \cdot (x + y), x + y, -y) \mid \\ x \in \mathcal{N}_L, u \cdot x = x, y \in \mathfrak{t}_d + \mathfrak{n}_P, t \in T_d, u \in \mathcal{U}_L, v \in U_P\}.$$

Note that $(x + y) - tuv \cdot (x + y) \equiv x - u \cdot x \pmod{\mathfrak{n}_P}$. Take $P_d := T_d \mathcal{U}_L U_P$ and $\mathfrak{p}_d := \mathfrak{t}_d + \mathcal{N}_L + \mathfrak{n}_P$, and let $pr : G/P \times G \times A_d \rightarrow G \times A_d$ be the projection. Then

$$\rho_{pr}^{-1}(SS(i_+^* \widetilde{\mathbb{M}}_a)) \cong G \times^P \Lambda_P,$$

where

$$\Lambda_P = \{(h, q_P(h), z, -\bar{z}) \in P_d \times A_d \times \mathfrak{p}_d \times \mathfrak{t}_d \mid h \cdot z = z\},$$

where we write $\bar{z}$ for the projection of z to $\mathfrak{t}_d \cong \mathfrak{p}_d/[\mathfrak{p}_d, \mathfrak{p}_d]$. Therefore, $SS(\mathbb{M}_a) \subseteq \overline{G \cdot \Lambda_P}$.

We claim that $\overline{G \cdot \Lambda_P} = \overline{G \cdot \Lambda_L}$. The inclusion $\Lambda_L \subseteq \Lambda_P$ is immediate. Conversely, take $(h, q_P(h), z, -\bar{z}) \in \Lambda_P$. Suppose that the commuting pair (h, z) has distinct joint eigenvalue pairs $(\xi_i, \eta_i)_{i=1}^k$, with corresponding generalized eigenspaces of dimensions rm_i , where $\sum_{i=1}^k m_i = d$. After replacing (h, z) by a G -conjugate, we may assume that

$$(h, z) \in \left(\prod_{i=1}^k \mathrm{GL}_{rm_i} \right) \times \left(\prod_{i=1}^k \mathfrak{gl}_{rm_i} \right).$$

Let h_i and z_i be the i -th blocks of h and z , respectively. Then $(\xi_i^{-1}h_i - I, z_i - \eta_i I)$ is a pair of commuting nilpotent matrices.

By [Ba], the variety of pairs of commuting nilpotent matrices is irreducible, and the open locus on which the first matrix is regular nilpotent is dense. To show $\Lambda_P \subset \overline{G \cdot \Lambda_L}$, it therefore suffices to consider

$$(\xi_i^{-1}h_i - I, z_i - \eta_i I) = (J_{rm_i}, p_i(J_{rm_i})),$$

where J_{rm_i} is a regular nilpotent Jordan block and p_i is a polynomial satisfying $p_i(0) = 0$.

The matrix J_{rm_i} lies in the closure of the regular locus with m_i distinct eigenvalues and one Jordan block of size r over each eigenvalue. Deforming J_{rm_i} within this locus and simultaneously deforming $p_i(J_{rm_i})$ by applying the same polynomial, we obtain commuting pairs lying in $G \cdot \Lambda_L$. Hence $\Lambda_P \subseteq \overline{G \cdot \Lambda_L}$. Together with the reverse inclusion, this proves $\overline{G \cdot \Lambda_P} = \overline{G \cdot \Lambda_L} (= \Lambda)$. Consequently, $SS(\mathbb{M}_a) \subseteq \Lambda$.

Now

$$\Lambda_L \cong T_d \times \mathfrak{t}_d \times \{(u, v) \in \mathcal{U}_L \times \mathcal{N}_L \mid u \cdot v = v\}.$$

The map $u \mapsto u - I$ identifies the last factor with a product of varieties of pairs of commuting nilpotent matrices. Therefore Λ_L is irreducible by [Ba], and it follows that $\Lambda = \overline{G \cdot \Lambda_L}$ is irreducible.

Finally, on the dense smooth locus $\widetilde{G}_d^\circ \subset G \times A_d$, the restriction of $\mathbb{M}_a$ is a nonzero integrable connection. Therefore $T_{\widetilde{G}_d^\circ}^*(G \times A_d) \subseteq SS(\mathbb{M}_a)$. At a point $(tu, \nu(t)) \in \widetilde{G}_d^\circ$, where $t \in T_d^\circ$ and $u \in O_{(r)}^d$, we have

$$T_{\widetilde{G}_d^\circ, (tu, \nu(t))}^*(G \times A_d) = \{(x + y, -y) \mid x \in \mathcal{N}_L, u \cdot x = x, y \in \mathfrak{t}_d\}.$$

This is precisely the restriction of Λ to the dense locus where $t \in T_d^\circ$ and every component of u is regular unipotent. Hence $\Lambda = \overline{T_{\widetilde{G}_d^\circ}^*(G \times A_d)} \subseteq SS(\mathbb{M}_a)$. Combining this with the

reverse inclusion proves $SS(\mathbb{M}_a) = \Lambda$. Moreover, Λ is Lagrangian because it is the closure of the conormal bundle to the smooth locally closed subvariety $\tilde{G}_d^\circ$. $\square$

For a connection $\mathcal{L}$ on a smooth locally closed subvariety $U \subset X$, we write $IC(X, \mathcal{L})$ for its minimal extension to X , supported on $\overline{U}$.

Let $\tilde{\mathbb{E}}_a^\circ$ denote the restriction of $\tilde{\mathbb{E}}_a$ to $\tilde{G}_d^\circ$.

Corollary 6.4. $\mathbb{M}_a$ is isomorphic to $IC(G \times A_d, \tilde{\mathbb{E}}_a^\circ)$.

Proof. For $i \neq 0$, the holonomic D -module $\mathcal{H}^i \mathbb{M}_a$ is supported on $\overline{\tilde{G}_d^\circ} \setminus \tilde{G}_d^\circ$. Its singular support is contained in a proper closed subvariety of Λ of dimension less than $\dim \Lambda$, by Lemma 6.3. On the other hand, the singular support of every nonzero holonomic D -module on X has pure dimension $\dim X$. Hence $\mathcal{H}^i \mathbb{M}_a = 0$ for all $i \neq 0$.

Set $M = \mathcal{H}^0 \mathbb{M}_a$. If K were a nonzero submodule or quotient of M supported on $\overline{\tilde{G}_d^\circ} \setminus \tilde{G}_d^\circ$, then $SS(K)$ would again be contained in a proper closed subvariety of Λ , resulting in the same contradiction as above. Thus M has no nonzero subobject or quotient supported on the boundary.

This is precisely the characterization of the intermediate extension, and the corollary follows. $\square$

Let $\mathbb{E}_a^\circ$ be the pushforward of $\tilde{\mathbb{E}}_a^\circ$ to G_d° along the S_d -étale map $q^\circ : \tilde{G}_d^\circ \rightarrow G_d^\circ$.

Proposition 6.5. We have $\overline{\mathbb{N}}_{a,r}^{\star d} = IC(G, \mathbb{E}_a^\circ)$.

Proof. The closure $\overline{\tilde{G}_d^\circ}$ inside $G \times A_d$ lies inside $G_d \times_{A_d/S_d} A_d$. Hence the projection $q : \overline{\tilde{G}_d^\circ} \rightarrow G_d$ is finite. Since finite pushforward is exact, it commutes with intermediate extension. Using Corollary 6.4, we then have

$$\overline{\mathbb{N}}_{a,r}^{\star d} = q_* IC(G \times A_d, \tilde{\mathbb{E}}_a^\circ) \cong IC(G, q_* \tilde{\mathbb{E}}_a^\circ) = IC(G, \mathbb{E}_a^\circ). \quad \square$$

Proof of Theorem 6.1. We first show there is an injection $\mathbb{C}[S_d] \hookrightarrow \text{End}_{\text{VHS}}(\mathbb{E}_a^\circ)$. Let $(\tilde{\mathbb{E}}_a^\circ)^e$ be the connection on $T_d^\circ \cdot O_{(r)}^d$ pulled back from the connection $\mathbb{K}_a^{\boxtimes d}$ on $O_{(r)}^d$. Then as discussed earlier, $(\tilde{\mathbb{E}}_a^\circ)^e$ is $S(L)$ -equivariant. Note that $\tilde{G}_d^\circ \cong G \times^L (T_d^\circ \cdot O_{(r)}^d)$. Hence we have $\tilde{\mathbb{E}}_a^\circ = \text{Ind}_{S(L)}^{\text{SL}_n} (\tilde{\mathbb{E}}_a^\circ)^e$. Now $(\tilde{\mathbb{E}}_a^\circ)^e$ is equivariant under the permutation S_d -action on the d -blocks. Let $N(S(L))$ be the normalizer of $S(L)$ in SL_n . Then $(\tilde{\mathbb{E}}_a^\circ)^e$ is further $N(S(L))$ -equivariant. Therefore, $\tilde{\mathbb{E}}_a^\circ$ is S_d -equivariant for the deck action. Hence S_d acts on $\mathbb{E}_a^\circ$. Since S_d acts on every stalk of $\mathbb{E}_a^\circ$ via its regular action, the action is faithful and we obtain the embedding $\mathbb{C}[S_d] \hookrightarrow \text{End}_{\text{VHS}}(\mathbb{E}_a^\circ)$.

Now we use Proposition 6.5. Since intermediate extension is fully faithful, the theorem follows. $\square$

For any $\tau \vdash d$, let V_τ be the corresponding irreducible module of S_d . Let $\mathbb{E}_{a,\tau} = \text{Hom}_{S_d}(V_\tau, \mathbb{E}_a^\circ)$. Then, under the S_d -action, $\overline{\mathbb{N}}_{a,r}^{\star d}$ decomposes as

$$\overline{\mathbb{N}}_{a,r}^{\star d} \cong \bigoplus_{\tau \vdash d} V_\tau \otimes IC(G, \mathbb{E}_{a,\tau}),$$

where each summand is viewed on G by closed pushforward.

Corollary 6.6. *Under the S_d -action from Theorem 6.1, $\mathrm{gr}(\overline{\mathbb{N}}_{a,r}^{\star d})$ decomposes as*

$$\mathrm{gr}(\overline{\mathbb{N}}_{a,r}^{\star d}) \cong \bigoplus_{\tau \vdash d} V_\tau \otimes \mathrm{gr}(IC(G, \mathbb{E}_{a,\tau})).$$

6.2. Lie algebra case. Let $\mathfrak{g} = \mathfrak{gl}_n$. The notations $\mathfrak{g}_d^\circ$, $\widetilde{\mathfrak{g}}_d^\circ$, $\mathbb{E}_{a,\tau}$, etc. have meaning similar to the corresponding notations for G .

There is an analogue of Corollary 6.6 in this setting, which answers the first part of [GN2, Conjecture 1.2] replacing their $\mathcal{H}_{m,n}$ by the derived version of their $\mathcal{H}'_{m,n}$ (these sheaves are conjecturally the same up to a grading shift, cf. [GN2, Conjecture 2.2]). Moreover, by [Ma, Proposition 3.8], $\mathcal{H}'_{m,n}$ is the Hodge associated graded of $\overline{\mathbb{N}}_m$ if $\mathrm{GCD}(m, n) = 1$.

The description of the decomposition can be made more explicit in the style of [HK] and [?]. Let $\mathfrak{F}$ denote the Fourier transform on $\mathcal{D}_{\mathfrak{g}}$ -modules. For any $\tau = (\tau_i) \vdash d$, let $O_{r\tau} \subset \mathfrak{g}$ denote the nilpotent orbit corresponding to $r\tau := (r\tau_1, \dots, r\tau_k)$. Let $b = \mathrm{gcd}(\tau_i)$. Then $\pi_1(O_{r\tau}) = \pi_0(\mathrm{Stab}_{\mathrm{SL}_n}(x)) = \mathbb{Z}/br\mathbb{Z}$ for an element $x \in O_{r\tau}$. For an integer a coprime to r , let $\mathbb{K}_{a,\tau}$ be the integrable connection on $O_{r\tau}$ corresponding to $e^{2\pi a\sqrt{-1}/r} \in (\mathbb{Z}/br\mathbb{Z})^*$. Let τ^t denote the transpose of τ .

Proposition 6.7. $\mathfrak{F}(IC(\mathfrak{g}, \mathbb{E}_{a,\tau})) = IC(\mathfrak{g}, \mathbb{K}_{a,\tau^t})$.

Proof. First consider the case when $d = 1$. Here let $\mathbb{N}_a$ denote the $\mathcal{D}_{\mathrm{sl}_n}$ -module analogous to the group version in Section 4.2. Under the decomposition $\mathfrak{gl}_n = \mathfrak{sl}_n \oplus \mathfrak{z}$, we have $IC(\mathfrak{g}, \mathbb{E}_{a,(1)}) = \mathbb{N}_a \boxtimes \mathcal{O}_{\mathfrak{z}}$. Let z be the coordinate of $\mathfrak{z}$ and $\delta_{\mathfrak{z}} := \mathcal{D}_{\mathfrak{z}}/\mathcal{D}_{\mathfrak{z}}z$. Then $IC(\mathfrak{g}, \mathbb{K}_{a,(1)}) = \mathbb{N}_a \boxtimes \delta_{\mathfrak{z}}$. Let $\mathfrak{F}_{\mathrm{sl}_n}$ denote the Fourier transform for $\mathcal{D}_{\mathrm{sl}_n}$ -modules. Then the proposition in this case is implied by the isomorphism $\mathfrak{F}_{\mathrm{sl}_n}(\mathbb{N}_a) \cong \mathbb{N}_a$. This is shown in the proof of [BeLo, Proposition 6.4].

We proceed to the case $d > 1$. Let $\mathfrak{n}_{\mathfrak{p}}$ be the nilpotent radical of $\mathfrak{p}$ with Levi decomposition $\mathfrak{p} = \mathfrak{l} \ltimes \mathfrak{n}_{\mathfrak{p}}$. Let $O_{(r)} \subset \mathfrak{gl}_r$ be the regular nilpotent orbit. Define $\widetilde{\mathfrak{g}}_d := G \times^P (\mathfrak{t}_d + O_{(r)}^d + \mathfrak{n}_{\mathfrak{p}})$ with $\pi_1(\widetilde{\mathfrak{g}}_d) = (\mathbb{Z}/r\mathbb{Z})^d$. Let $\widetilde{\mathcal{L}}_a$ be the integrable connection on $\widetilde{\mathfrak{g}}_d$ corresponding to the diagonal image of $e^{2\pi a\sqrt{-1}/r}$ in $((\mathbb{Z}/r\mathbb{Z})^*)^d$. Let $p : G \times^P \mathfrak{p} \rightarrow \mathfrak{g}$ be the action map and $j : \widetilde{\mathfrak{g}}_d \rightarrow G \times^P \mathfrak{p}$ be the embedding. By definition, $\overline{\mathbb{N}}_{a,(r)}^{\star d} = p_{\dagger} j_{\dagger} \widetilde{\mathbb{E}}_a$.

Let $\widetilde{\mathbb{K}}_a$ be the restriction of $\widetilde{\mathbb{E}}_a$ to $G \times^P (O_{(r)}^d + \mathfrak{n}_{\mathfrak{p}})$. Let $\mathfrak{F}_{\mathfrak{g}}$ denote the partial Fourier transform on $G \times^P \mathfrak{g}$ along the $\mathfrak{g}$ direction. Note that $(\mathfrak{g}/\mathfrak{p})^* \cong \mathfrak{n}_{\mathfrak{p}}$ under the trace form. This plus the proposition in the case of $d = 1$ implies that $\mathfrak{F}_{\mathfrak{g}}(IC(G \times^P \mathfrak{g}, \widetilde{\mathbb{E}}_a)) = IC(G \times^P \mathfrak{g}, \widetilde{\mathbb{K}}_a)$ (the same holds if we replace the intermediate extensions with the $\dagger$ - or $!$ -extensions in both parts). By Springer theory, there is an S_d action on $q_{\dagger} IC(G \times^P \mathfrak{g}, \widetilde{\mathbb{K}}_a)$ for $q : G \times^P \mathfrak{g} \rightarrow \mathfrak{g}$. Similar to [HK, Theorem 5.2], there is an isomorphism of S_d -modules:

$$\mathfrak{F} q_{\dagger}(IC(G \times^P \mathfrak{g}, \widetilde{\mathbb{E}}_a)) = q_{\dagger} IC(G \times^P \mathfrak{g}, \widetilde{\mathbb{K}}_a) \otimes \mathrm{sign}.$$

As $V_\tau \otimes \mathrm{sign} \cong V_{\tau^t}$, it remains to show the S_d -module decomposition

$$q_{\dagger} IC(G \times^P \mathfrak{g}, \widetilde{\mathbb{K}}_a) \cong \bigoplus_{\tau \vdash d} V_\tau \otimes IC(\mathfrak{g}, \mathbb{K}_{a,\tau}).$$

For this, we apply the generalized Springer correspondence [LS, Proposition 5.2] for the cuspidal datum

$$(L \cap \mathrm{SL}_n, O_{(r)}^d, \mathcal{F}_a^{\boxtimes d}),$$

which has relative Weyl group S_d . It identifies the multiplicity space attached to $(O_{r\tau}, \mathbb{K}_{a,\tau})$ with the irreducible representation V_τ . $\square$

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